LITERATURE REVIEW

Neuro-Symbolic AI

Foundations, canonical works, and the state of the art through mid-2026

Prepared: August 2026

Coverage: 1943 – August 2026

Written for: a data scientist with a strong ML background and no prior exposure to the field

> **On citations.** Every reference was checked against a bibliographic source (arXiv API, OpenAlex, dblp, publisher pages). Roughly 75 citations were independently adjudicated in a dedicated audit pass: **zero fabricated papers, zero invented first authors, zero invented venues** were found; twelve metadata errors were, and have been corrected in place.
>
> Anything that could *not* be confirmed carries a ⚠ and states why. Treat those as leads to verify, not as established facts.

Contents

Part II — Annotated Bibliography

Neuro-Symbolic AI: A Literature Review for the Entering Researcher

Written for a data scientist with a strong ML background and no prior exposure to the field. Literature covered through August 2026.

0. How to read this document

This review is organised so you can stop at any depth. Section 1–2 give you the field's history and its vocabulary — enough to read any NeSy paper's introduction. Section 3 is the technical core: eight families of method, each with its central mechanism, its canonical papers, and the precise place it breaks. Section 4 is what changed in 2025–26. Section 5 is the sceptical audit — read it before you believe anything in Section 3. Sections 6–9 are practical: open problems, a reading path, runnable tooling.

On citations. Every reference here was checked against a bibliographic source (arXiv API, OpenAlex, dblp, publisher pages) during preparation. Where a citation or a number could *not* be confirmed, it carries a ⚠ and says why. Twelve metadata errors found during the audit have been corrected in place; a few widely-repeated figures turned out to be unsupported and have been dropped rather than repeated. Full annotated bibliography in Part II — Annotated Bibliography.

A warning about the field's own claims. Neuro-symbolic AI has an unusually large gap between what its abstracts assert and what its experiments show. This is not fraud — the community documents the gap itself, often better than its critics do — but a newcomer who reads only the positive literature will form a badly calibrated picture. Section 5 exists for that reason and is not optional reading.

1. The problem the field exists to solve

1.1 The first neural network was a logic machine

McCulloch and Pitts' 1943 *A logical calculus of the ideas immanent in nervous activity* (Bull. Math. Biophysics 5:115–133) modelled the neuron as a threshold logic unit and showed that networks of them realise arbitrary propositional logic. This matters rhetorically: neuro-symbolic researchers frame their programme as *reunification*, not hybridisation. The two traditions had a common ancestor and drifted.

1.2 The systematicity dispute

The drift hardened into philosophy. Fodor and Pylyshyn's *Connectionism and cognitive architecture: A critical analysis* (Cognition 28:3–71, 1988) argued that thought is *systematic* — anyone who understands "John loves Mary" understands "Mary loves John" — and that a network can display systematicity only by implementing a classical symbol system, in which case it is an implementation detail rather than a cognitive theory.

Smolensky answered constructively. *Tensor Product Variable Binding and the Representation of Symbolic Structures in Connectionist Systems* (Artificial Intelligence 46(1–2):159–216, 1990) gave an algebra — role and filler vectors bound by outer product and superposed — in which exact, invertible variable binding lives entirely inside a vector space. Every modern vector-symbolic architecture descends from it, and so does a good deal of current work on binding in transformers.

This dispute has since been adjudicated empirically, and the review literature usually forgets to say so. Lake and Baroni built SCAN (ICML 2018) to turn systematicity into a measurable benchmark; seq2seq models failed it badly. Then in *Human-like systematic generalization through a meta-learning neural network* (Nature 623(7985):115–121, 2023) the same authors showed that a *standard transformer*, meta-trained over a stream of freshly generated compositional tasks, matches human systematicity head-to-head — no symbolic machinery at all. The lesson is not that Fodor was wrong but that the requirement can be met by changing the training distribution rather than the architecture. Any argument that systematicity *requires* symbols has to answer this paper.

1.3 The engineering line

Running in parallel and largely ignoring the philosophy: Towell and Shavlik's KBANN (Artificial Intelligence 70(1–2):119–165, 1994) compiled a propositional domain theory directly into a network's topology and initial weights, then refined it by backpropagation. Every later "knowledge injection" method is a variation on KBANN.

Valiant gave the programme institutional legitimacy: *Robust logics* (Artificial Intelligence 117(2):231–253, 2000) and *Three problems in computer science* (JACM 50(1):96–99, 2003) named the reconciliation of statistical learning with logical reasoning as a foundational open problem in computing. This is the most-cited legitimacy argument in the field's introductions.

1.4 The third wave

The modern re-founding is largely Artur d'Avila Garcez and Luís Lamb's: *Neural-Symbolic Cognitive Reasoning* (Springer, 2009) systematised neural encodings of modal, temporal and epistemic logics; *Neurosymbolic AI: The 3rd Wave* (arXiv 2012.05876, 2020; AI Review 2023) supplied the field's self-narrative and the vocabulary everyone now uses.

1.5 The competing programmes

You need these to understand what NeSy is arguing *against*:

Position	Core claim	Anchor
Marcus	Scale alone will not produce robustness; need explicit cognitive models and symbol manipulation	The Next Decade in AI (2020)
Bengio (System 2)	Keep everything differentiable, but import symbolic desiderata as *inductive biases* — sparse causal mechanisms, attention as dynamic binding	Goyal & Bengio, Proc. R. Soc. A 478:20210068 (2022)
LeCun	Reject both; reasoning is energy-based inference over a learned world model	A Path Towards Autonomous Machine Intelligence (2022)
Chollet	Reframe as *measurement*: intelligence is skill-acquisition efficiency relative to priors	On the Measure of Intelligence (2019) → ARC
Sutton (bitter lesson)	Hand-crafted structure loses to general methods plus compute	2019 blog post ⚠ *(no formal venue)*

The most serious NeSy reply to Sutton is Velasquez, **Neel Bhatt**, Topcu, Wang, Sycara, Stepputtis, Neema & Vallabha, *Neurosymbolic AI as an antithesis to scaling laws* (PNAS Nexus 4(5):pgaf117, 2025). It concedes the history and argues that using prior knowledge and discovering new knowledge are not exclusive. Read it — but read it noticing that its efficiency figures ("100× smaller than GPT-3", "96.9% fewer parameters") are the authors' summaries of scattered single-domain case studies, not general results. The audit traced them: the 100× figure is West et al. 2022 on symbolic knowledge distillation; the 96.9% is Zhao et al. 2024 on physics-informed autonomous driving.

2. How the field maps itself

2.1 Kautz's six types

The organising device newcomers meet first. Presented in Henry Kautz's AAAI-2020 Engelmore Memorial Lecture, published as *The Third AI Summer* (AI Magazine 43(1):93–104, 2022), and transmitted in print by Sarker, Zhou, Eberhart & Hitzler, *Neuro-symbolic artificial intelligence* (AI Communications 34(3):197–209, 2021).

Ordered by tightness of coupling:

1. **symbolic Neuro symbolic** — a network whose inputs and outputs are symbols (BERT/GPT over subword tokens)
2. **Symbolic[Neuro]** — a symbolic algorithm calling a neural subroutine (AlphaGo: MCTS with a learned evaluator)
3. **Neuro;Symbolic** — neural perception emits symbols consumed by a separate reasoner (NS-CL; LLM → theorem prover)
4. **Neuro:Symbolic→Neuro** — symbolic machinery generates training data for a network
5. **Neuro_{Symbolic}** — logic compiled into the network or softened into differentiable constraints (KBANN, Logic Tensor Networks)
6. **Neuro[Symbolic]** — a symbolic reasoning engine embedded inside the neural engine

Types 1 and 2 describe almost everything deployed. Types 5 and 6 are what the founders wanted. Kautz himself notes no genuine Type 6 system exists.

> ⚠ **Citation hygiene.** The taxonomy could not be verified against Kautz's printed text (the Wiley DOI returns HTTP 402; the AAAI OJS article page errors). The category names and examples above come from Sarker et al. 2021 and secondary sources. **Cite Sarker et al. 2021 for the taxonomy, not Kautz.** Notation also varies across secondary sources (`Neuro;Symbolic` vs `Neural | Symbolic`; `Neuro_{Symbolic}` vs `NEUROSYMBOLIC`) — don't quote a rendering as canonical.

2.2 The competitor nobody cites

The taxonomy is *contested*, and reviews that present Kautz as settled are making an interpretive move. Van Harmelen & ten Teije, *A Boxology of Design Patterns for Hybrid Learning and Reasoning Systems* (Journal of Web Engineering 18(1–3):97–124, 2019) gives a *compositional* grammar — boxes for models, actors, data and symbols, wired into nestable patterns — where Kautz gives a single ordinal scale. Read both.

2.3 The map that will actually help you

For a reader with strong ML background, the best technical map is Marra, Dumančić, Manhaeve & De Raedt, *From statistical relational to neurosymbolic artificial intelligence: A survey* (Artificial Intelligence 328:104062, 2024). It positions NeSy against 25 years of statistical relational AI along seven dimensions — semantics (proof-based vs model-based), fuzzy vs probabilistic, directed vs undirected, type of logic, parameter vs structure learning, symbolic vs subsymbolic representation, and how logic is grounded. It explains *why* DeepProbLog, Logic Tensor Networks, Scallop and semantic loss differ, in terms of inference and semantics rather than block diagrams. It also quietly demonstrates that many "new" NeSy systems are re-derivations of StarAI ideas.

Two more worth knowing: Wang, Yang & Wu, *Towards Data- and Knowledge-Driven AI: A Survey on Neuro-Symbolic Computing* (IEEE TPAMI 2024) — the highest-visibility non-Western synthesis, which partitions the field differently and covers the perception-heavy literature the European probabilistic-logic surveys underweight. And Hitzler, Sarker & Eberhart (eds.), *Compendium of Neurosymbolic Artificial Intelligence* (IOS Press, FAIA vol. 369, 2023) — 24 invited chapters, the closest thing to a handbook. *(Not open access. The "704 pp., 30 chapters" figure circulating online is wrong; the publisher page shows 24 chapters running to page 546.)*

2.4 The LLM-era split

Since 2023 the useful vocabulary is Yang, Shao, Guo, Zhang, Zhou, Jia, Dai & Li, *Neuro-Symbolic AI: Towards Improving the Reasoning Abilities of Large Language Models* (IJCAI 2025 Survey Track), which splits the post-2023 literature three ways:

- **Symbolic→LLM** — symbolic structure shapes prompting or training
- **LLM→Symbolic** — the LLM formalises, a solver decides
- **LLM+Symbolic** — interleaved loops

Most 2026 papers assume this vocabulary. It is a taxonomy, not a controlled comparison — don't cite it for results.

2.5 There is no agreed formal definition

De Smet and De Raedt's *Defining neurosymbolic AI* (2025) proposes defining neurosymbolic inference as the computation of an integral over a product of a logical function and a belief function. Whether that covers embedding-based, LTN-style and LLM+solver systems — or quietly excludes them — is unsettled. In practice "neurosymbolic" in 2026 is a family resemblance, not a definition, and the RAIL position paper (§4.5) makes the boundary broader still.

---

3. The eight technical families

Each subsection follows the same shape: the mechanism in one paragraph, the papers, and where it breaks.

3.1 Logic as a loss — fuzzy and differentiable logic

The mechanism. Classical logic is discrete, so you can't backpropagate through it. Replace truth values {0,1} with [0,1]; replace AND by a t-norm (Gödel/min, Łukasiewicz/max(0,a+b−1), or product a·b), OR by its dual, implication by a residuum or S-implication, and the quantifiers by aggregators. Let the atoms be network outputs — this mapping from logical symbols to tensors is called **grounding** — and a first-order formula becomes an ordinary differentiable computation graph emitting a satisfaction score. Convert to a loss, add to cross-entropy, done. Cost is roughly linear in formula size × batch size. No probabilistic inference, no model counting, no combinatorial blow-up. **This is the standard on-ramp to the field.**

The arc. Semantic-Based Regularization (Diligenti, Gori & Saccà, Artificial Intelligence 244:143–165, online 2015 / print 2017) formalised "constraints as regularisers". **Logic Tensor Networks** — Serafini & d'Avila Garcez's 2016 proposal, matured in Badreddine, d'Avila Garcez, Serafini & Spranger's 68-page AIJ 303:103649 (2022) — gave the reference formulation: *Real Logic*, in which constants ground to tensors, functions and predicates to networks, and learning maximises satisfiability of a knowledge base. LYRICS (Marra et al., ECML-PKDD 2019) generalised SBR and LTN into a declarative front-end. DL2 (Fischer et al., ICML 2019) abandoned fuzzy semantics entirely, translating constraints over real-valued network quantities into a loss that is exactly zero when satisfied — and reused the same machinery to *query* a trained net by gradient search. IBM's Logical

Neural Networks (Riegel, Gray, Luus et al., 2020) went furthest: each neuron *is* a weighted real-valued connective carrying truth-value **bounds**, giving open-world semantics and omnidirectional inference.

A parallel branch relaxes structure rather than constraints. Neural LP (Yang, Yang & Cohen, NIPS 2017) turns chain rules over a knowledge graph into attention-weighted sparse matrix products. dILP (Evans & Grefenstette, JAIR 61:1–64, 2018) enumerates rule templates and learns continuous weights over candidate clauses, buying the noise robustness classical ILP lacks. Neural Theorem Provers (Rocktäschel & Riedel, NIPS 2017) unroll Prolog backward chaining and replace unification with an RBF kernel on symbol embeddings — beautiful, but the proof tree explodes; Minervini, Riedel, Stenetorp, Grefenstette & Rocktäschel's *Learning Reasoning Strategies in End-to-End Differentiable Proving* (ICML 2020) learns a rule-selection strategy to make it tractable. *(Note: "Conditional Theorem Provers" is the method name, not the paper title — a very common miscitation.)*

A third branch puts logic in the architecture. difflogic (Petersen, Borgelt, Kuehne & Deussen, NeurIPS 2022) parameterises each neuron as a relaxed softmax over the 16 binary Boolean gates, trains by gradient descent, then discretises to hard gates. Convolutional difflogic (NeurIPS 2024 Oral) reaches **86.29% on CIFAR-10 with 61 million logic gates, 29× smaller than prior SOTA**. *(The widely-circulated "4 nanosecond inference" and "29×–61×" figures are not in the paper — they came from a social-media post. Drop them.)*

Where it breaks — and this is the most valuable thing in the sub-area.

Operator pathologies. Van Krieken, Acar & van Harmelen, *Analyzing Differentiable Fuzzy Logic Operators* (Artificial Intelligence 302:103602, 2022) differentiated the whole operator zoo and found most unusable: Gödel min/max routes gradient to a single argument; Łukasiewicz operators have large zero-gradient plateaus; product t-norms vanish as operands saturate; and residuated implications have a severe antecedent/consequent gradient imbalance, so the cheapest way to "satisfy" A→B is to push A to zero and make it vacuously true. **Their headline finding: the configurations that train best are non-standard combinations that no longer obey ordinary logical laws.** You are optimising a logic-*shaped* surrogate, not logic.

Reasoning shortcuts. Even with a perfect relaxation, the constraint under-determines the concepts. Marconato, Teso, Vergari & Passerini, *Not All Neuro-Symbolic Concepts Are Created Equal* (NeurIPS 2023), showed a model can hit 100% task accuracy and full constraint satisfaction while grounding symbols with entirely wrong semantics — an identifiability failure. This broke the field's central marketing claim: *satisfying the constraint does not mean the network learned the right symbols*, so the promised interpretability and OOD robustness do not follow.

> **Priority note.** The reasoning-shortcut literature is usually dated to 2023. It shouldn't be. Topan, Rolnick & Si, *Techniques for Symbol Grounding with SATNet* (2021), showed that SATNet's celebrated visual-Sudoku result depended on **label leakage** and that SATNet fails at symbol grounding without extra machinery. That is the same failure, documented two years earlier, on a headline result. It is also the field's cleanest replication failure and belongs in every discussion of NeSy evaluation.

The 2026 synthesis is Marconato, Bortolotti, van Krieken, Morettin, Umili, Vergari, Tsamoura, Passerini & Teso, *Symbol Grounding in Neuro-Symbolic AI: A Gentle Introduction to Reasoning Shortcuts* (JAIR, special track on Integration of Logical Constraints in Deep Learning).

Status in 2026. Almost no new *frameworks* since ~2022; the intellectual energy since 2023 has gone into the critique literature — a field auditing itself. Two branches are healthy for opposite reasons: differentiable logic-gate networks (an efficiency story, not a reasoning story, with a real niche in ultra-low-latency and hardware inference), and the reasoning-shortcut/identifiability line, which now has a benchmark suite (rsbench), complexity results, and a JAIR synthesis.

> ⚠ **A verdict I have softened.** One research thread concluded that differentiable ILP (dILP, NTP/CTP, Neural LP, DRUM, RNNLogic) "looks substantially superseded". That is a conclusion about one sub-community. Counter-evidence from an entirely separate cluster (§3.3): Gao, Inoue et al., *Differentiable Rule Induction from Raw Sequence Inputs* (ICLR 2025); Eiter, Inoue & Moriyama, *Neural Decision-Propagation for Answer Set Programming* (IJCAI-ECAI 2026, arXiv:2605.01797); Baugh et al., *Disentangling Neural Disjunctive Normal Form Models* (NeSy 2025); Rader & Russo on vectorised NeurASP (TPLP). The line is smaller than it was, not dead.

3.2 Logic as exact inference — probabilistic NeSy, weighted model counting, tractable circuits

The whole sub-area rests on one reduction. Take a logic program whose facts are independent Bernoulli variables, ask for the probability that a query succeeds, and you get **weighted model counting (WMC)**: sum the weights of all satisfying assignments of a Boolean formula. Fierens et al., *Inference and learning in probabilistic logic programs using weighted Boolean formulas* (TPLP 15(3):358–401, 2015), made this the standard ProbLog pipeline — ground, collect proofs into a provenance formula, compile that into a circuit, evaluate.

WMC is #P-hard. Everything downstream is one of three things, and if you internalise only one thing from this section, make it this trichotomy:

1. **Compilation** — pay an exponential once, evaluate cheaply forever
2. **Approximation** — drop proofs, sample, or learn a surrogate for the count
3. **Restriction** — pick a logical fragment where counting is easy

Why compilation makes the neural part possible. Darwiche & Marquis' *A Knowledge Compilation Map* (JAIR 17:229–264, 2002) organises target languages by succinctness vs which queries they answer in polytime. Two structural properties make counting linear: *decomposability* (an AND's children share no variables, so products factorise) and *determinism* (an OR's children are mutually exclusive, so sums don't double-count). A compiled d-DNNF or SDD is literally an arithmetic circuit of sums and products — **and an arithmetic circuit is differentiable.** That is the hinge the entire construction turns on. Probabilistic circuits (sum-product networks; PSDDs) are the same object viewed as a learnable model class.

The systems. DeepProbLog (Manhaeve, Dumančić, Kimmig, Demeester & De Raedt, NeurIPS 2018; journal version Artificial Intelligence 298, article 103504, 2021) introduced the **neural predicate**: the network's softmax supplies fact probabilities, the compiled circuit computes the query probability, gradients flow back through sums and products. **MNIST-addition** — two digit images, supervision only on their sum — became the field's fruit fly because it isolates the hard part exactly: latent discrete concepts supervised only through a symbolic function.

Then: NeurASP (Yang, Ishay & Lee, IJCAI 2020) swaps Prolog for answer set programming, which is far more natural for combinatorial constraints. SLASH backs the neural predicate with a probabilistic circuit. DeepStochLog (AAAI 2022) restricts to stochastic definite clause grammars, turning inference into parsing. NeuPSL (IJCAI 2023) abandons discrete counting for a convex energy over continuous truth values. Embed2Sym (KR 2022) skips differentiable reasoning entirely — cluster the embeddings, then label the clusters with a solver. **Scallop** (Huang, Li, Chen, Samel, Naik, Song & Si, NeurIPS 2021; Li, Huang & Naik, PACMPL 7(PLDI):1463–1487, 2023) generalises all of it with a **provenance semiring**, where the differentiable top-k-proofs semiring gives an explicit dial trading exactness for cost.

Xu, Zhang, Friedman, Liang & Van den Broeck's semantic loss (ICML 2018) is the same WMC machinery used as a *regulariser* rather than an architecture — the cheapest possible entry point for an ML practitioner: one extra loss term, no architecture change, no inference-time cost.

Why Markov logic networks lost. Richardson & Domingos' MLNs (Machine Learning 62(1–2):107–136, 2006) are the historical precursor: weighted first-order formulas defining a ground Markov network. They lost because an undirected model needs a global partition function over an exploding grounding, and there was no compact differentiable computation graph to hand to autograd. Directed probabilistic-logic semantics plus knowledge compilation give you that graph for free. This is why the neural wave built on ProbLog rather than Alchemy.

Since 2023 the story is almost entirely scaling. A-NeSI (van Krieken et al., NeurIPS 2023) replaces the counting step with learned neural surrogates, pushing MNIST-addition from ~4 digits to 15. On the systems side: KLay (ICLR 2025) restructures irregularly sparse circuits into GPU-parallel "knowledge layers"; Dolphin (ICML 2025) vectorises provenance on GPU, reporting **1.71×–62× speedups across 13 benchmarks** and convergence on hard benchmarks where Scallop, ISED and IndeCateR+ fail entirely; DeepLog compiles to GPU algebraic circuits; REASON (HPCA 2026) is a dedicated accelerator. **This shift from new formalisms to systems engineering is what a field does when its conceptual core is settled — not what a dying field does.**

In parallel the field turned self-critical. Van Krieken, Minervini, Ponti & Vergari, *On the Independence Assumption in Neurosymbolic Learning* (ICML 2024), prove that the near-universal assumption that symbol probabilities are conditionally independent given the input biases these systems toward overconfidence, cannot represent uncertainty over multiple valid assignments, and yields highly disconnected non-convex minima. **This is the most important negative result here: independence is what makes WMC factorise and therefore what makes the whole approach tractable — so the tractability trick has a real statistical cost, not just an approximation cost.** The constructive answer is Neurosymbolic Diffusion Models (NeurIPS 2025): discrete diffusion over the symbol sequence, reusing independence only *within* each step.

Honest scale. A-NeSI's own comparison table: DeepProbLog 97.20% at N=1 digit, 95.20% at N=2, **timeout at N=4**; DeepStochLog times out at N=15; A-NeSI reaches N=15 at **75.90% ± 2.21**. The benchmark that defines the field is simultaneously a toy and, once scaled, unsolved. There is no published demonstration of differentiable logical inference over a knowledge base of the size non-differentiable engines handle routinely.

3.3 Abduction and consistency — the family the Western surveys omit

Architecturally distinct, and the largest non-Western NeSy cluster. Zhi-Hua Zhou's Nanjing LAMDA group has run **Abductive Learning (ABL)** since 2018. Dai, Xu, Yu & Zhou, *Bridging Machine Learning and Logical Reasoning by Abductive Learning* (NeurIPS 2019), is the anchor.

The mechanism: a perception model proposes pseudo-labels; a symbolic knowledge base **abduces** the most consistent revision of those labels; the model is retrained on the revised labels; repeat. It optimises *consistency*, not a differentiable relaxation. **No fuzzy logic, no weighted model counting, no differentiability requirement at all.** Demonstrated on handwritten-equation decipherment where even the arithmetic operation is unknown.

Still producing: Hu, Dai, Jiang & Zhou (AAAI 2025 Oral, arXiv:2412.08457); *Curriculum Abductive Learning* (NeurIPS 2025, arXiv:2505.12275), which reports that search over abductive revisions is itself the bottleneck.

Why this matters for how you read the field. The standard claim that tightly-coupled NeSy is stuck at MNIST-plus-constraints scale is made almost entirely without engaging this line, and the standard framing that tight coupling *requires* differentiable relaxation or WMC is falsified by it. Nobody has run ABL, DeepProbLog and LTN on a common suite at matched supervision — that comparison is an obvious, unclaimed research contribution.

A related and equally invisible cluster is Japanese: Katsumi Inoue (NII), Taisuke Sato and Chiaki Sakama built a "logic programming in vector spaces" and differentiable-ASP line — Sato/Takemura/Inoue on end-to-end ASP computation; Gao, Inoue et al., *Differentiable Rule Induction from Raw Sequence Inputs* (ICLR 2025); Eiter, Inoue &

Moriyama, *Neural Decision-Propagation for Answer Set Programming* (IJCAI-ECAI 2026); Takemura & Inoue, *Differentiable Logic Programming to Mitigate Reasoning Shortcuts* (ICLP 2026, EPTCS 450:29–51).

3.4 Logic as a hard guarantee — constraint layers, constrained decoding, verification

The organising question. If you have a specification your model must never violate — a JSON schema, a class hierarchy, a physics equality, a safety envelope, a legal rule — *where do you put it?* There are exactly four places, and you should learn to read any paper in this area as an answer to that question.

(1) In the loss — soft, no guarantee

Semantic loss, DL2, t-norm relaxations. The cost is honest and severe: a differentiable, architecture-agnostic regulariser that improves data efficiency and often accuracy, and **guarantees nothing**. In non-convex settings the penalised and constrained problems are not equivalent — a fixed penalty silently converts a hard requirement into a trade-off, and tuning the coefficient changes the objective you are solving.

(2) In the architecture — hard at training time

- **C-HMCNN** (Giunchiglia & Lukasiewicz, NeurIPS 2020) — hierarchy constraints via a max-constraint module
- **MultiplexNet** (Hoernle, Karampatsis, Belle & Gal, AAAI 2022) — constraint in DNF, one output transformation per disjunct, 100% satisfaction of quantifier-free linear arithmetic. Its stated limitation (DNF blow-up) is exactly what the next entry fixes.
- **Semantic Probabilistic Layers** (Ahmed, Teso, Chang, Van den Broeck & Vergari, NeurIPS 2022) — multiply a compiled constraint circuit by an expressive probabilistic circuit, renormalise exactly. The product is itself tractable, so you keep a *calibrated distribution over the feasible set*. This is the reference guarantee layer.
- **DC3** (Donti, Rolnick & Kolter, ICLR 2021) — equality completion plus unrolled gradient correction for inequalities; the bridge to "learning to optimize"
- **HardNet** (Min & Azizan) — closed-form differentiable projection for input-dependent affine/convex constraints, **with a proof that universal approximation is retained**. This answers the standing objection that hard layers cost you capacity. ⚠ *arXiv only as of v4; no accepted venue listed.*

Missing from most NeSy reviews and probably the most relevant to your day job: the *predict-then-optimise* / decision-focused family. Vlastelica, Paulus, Musil, Martius & Rolínek, *Differentiation of Blackbox Combinatorial Solvers* (ICLR 2020), gives an informative gradient for *exact* combinatorial solvers (shortest path, TSP, min-cost matching) treated as black boxes, via implicit linear interpolation of the piecewise-constant solver output — one extra solver call per backward pass, no relaxation. This is where routing, matching and scheduling problems live, and where the Warcraft-shortest-path benchmark the NeSy literature keeps citing actually comes from. Its counterpart on the "learn the constraints instead of injecting them" side is SATNet (Wang, Donti, Wilder & Kolter, ICML 2019) — a differentiable MAXSAT layer — read together with the Topan et al. replication failure noted in §3.1.

(3) At decoding time — where NeSy actually shipped

For LLMs the constraint lives in a token mask. Geng, Josifoski, Peyrard & West, *Grammar-Constrained Decoding for Structured NLP Tasks without Finetuning* (EMNLP 2023), reframed "JSON mode" as a general symbolic interface. Willard & Louf's *Efficient Guided Generation for LLMs* (outlines) made it cheap by precomputing an FSM index over the vocabulary. XGrammar (MLSys 2025) split the vocabulary into context-independent tokens (~99%, precomputable into bitmasks) and context-dependent tokens (~1%, runtime pushdown-automaton stack), reporting **up to 100× over prior grammar-constrained decoding** with near-zero serving overhead. It is the default structured-generation backend for vLLM, SGLang and TensorRT-LLM.

This is, by deployment volume, the most widely used symbolic constraint on a neural model in the world. Whether it deserves the NeSy label is an interpretive question — but the mechanism is exactly type-2 Symbolic[Neuro].

Two gotchas you must know before shipping it:

- **Masking is not conditioning.** Park, Wang, Berg-Kirkpatrick, Polikarpova & D'Antoni, *Grammar-Aligned Decoding* (NeurIPS 2024), show greedy grammar-constrained decoding produces grammatical strings at probabilities **not** proportional to the model's own distribution conditioned on the grammar. Their ASAp provably converges to the correct one — at the cost of repeated sampling.
- **Over-narrow grammars destroy reasoning.** CRANE (Banerjee, Suresh, Ugare, Misailovic & Singh, ICML 2025) proves that a grammar admitting only syntactically valid *final answers* strips out the intermediate-computation strings the model needs, and recovers **up to 10 accuracy points** on GSM-Symbolic and FOLIO by augmenting the grammar with an unconstrained reasoning region. This is why modern schemas leave room for a scratchpad field.

(4) Outside the model — shields and verifiers

Shielding (Alshiekh, Bloem, Ehlers, Könighofer, Niekum & Topcu, AAAI 2018) synthesises a reactive system from an LTL safety spec that overrides unsafe actions, with analysis of when it preserves convergence. Probabilistic logic shields (Yang, Marra, Rens & De Raedt, IJCAI 2023, distinguished paper) make the shield differentiable via probabilistic logic programming so safety shapes the policy gradient rather than only vetoing at execution.

Formal verification proper runs Reluplex (Katz, Barrett, Dill, Julian & Kochenderfer, CAV 2017) → Marabou 2.0 (CAV 2024) → α,β-CROWN, which won VNN-COMP 2025. **Calibrate accordingly:** the guarantee is real, but the benchmark networks are small ONNX models and the properties are local robustness or control envelopes — not the transformer you are shipping. Claims that this "verifies AI systems" are overstated by roughly the gap between an ACAS Xu controller and a frontier model.

Practical takeaway, unglamorous but well-supported: use hard decoding constraints for format, hard layers or projections for numeric feasibility, external shields and checkers for safety, and treat soft losses as a data-efficiency trick — never as a guarantee.

3.5 Perception → symbols → executor

The engineering bet: deep nets are good at seeing and bad at long compositional inference, so split the job — have the network emit a structured, symbol-like description of the scene, and let a classical executor reason over it.

CLEVR (Johnson et al., CVPR 2017) made the bet testable. NS-VQA (Yi, Wu, Gan, Torralba, Kohli & Tenenbaum, NeurIPS 2018) ran Mask R-CNN plus an attribute CNN to build an explicit object table, an LSTM seq2seq to parse the question into a program, then executed the program on the table: **99.8% on CLEVR**. The transferable lesson is that symbolic execution is *length-robust* — accuracy does not decay as program depth grows. (Read the accuracy number critically; see §5.1.)

NS-CL (Mao, Gan, Kohli, Tenenbaum & Wu, ICLR 2019) removed most of the supervision: concepts became learned embeddings, execution became *quasi-symbolic* (soft masks over object slots) so gradients flow, and training used only (image, question, answer) triples. **This is the architectural template the field still uses: object-centric front-end → learned concept grounding → differentiable symbolic executor.**

Its direct ancestor, usually omitted, is Neural Module Networks (Andreas, Rohrbach, Darrell & Klein, 2015), which parses a question into a layout and dynamically assembles a network from reusable typed modules — and which stakes out the alternative position that the *executor itself* be neural.

Two probes followed. CLEVRER (ICLR 2020) added descriptive / explanatory / predictive / counterfactual questions over collision videos; NS-DR bolted a learned dynamics predictor between parser and executor. Its numbers are where the pipeline honestly breaks: **88.1% descriptive but 42.2% counterfactual.** CLEVR-Hans (Stammer, Schramowski & Kersting, CVPR 2021) deliberately confounds classes so a CNN passes validation and collapses on a de-confounded split — and shows that intervening on a symbolic scene representation ("never use colour") fixes it while pixel saliency cannot even *name* the bug. **That — semantic-level debuggability under confounding — is the real payoff of symbolic scene representations, not raw accuracy.**

The interpretability wing runs as concept bottleneck models (Koh et al., ICML 2020): predict human-named concepts, then predict the label only from them, which buys test-time intervention. Mahinpei, Clark, Lage, Doshi-Velez & Pan, *Promises and Pitfalls of Black-Box Concept Learning Models* (2021), showed the bottleneck **leaks**: soft/joint concept representations smuggle in information beyond the named concepts. The dispute is live in 2026, with work arguing both that CBMs never had an information bottleneck and that leakage is sometimes beneficial. There is still no agreed *measurement* of leakage.

On the perception side, Slot Attention (Locatello et al., NeurIPS 2020) is the canonical differentiable "K exchangeable slots" module. Scaling it to real photos required frozen self-supervised features (DINOSAUR, ICLR 2023), and slot count remains an a-priori hyperparameter. **There is no accepted method for extracting a reliable symbol table from an arbitrary photograph — which is what blocks this whole pipeline from leaving synthetic domains.**

Program induction, two branches. Bottom-up: DreamCoder (Ellis, Wong, Nye, Sablé-Meyer, Cary, Morales, Hewitt, Solar-Lezama & Tenenbaum, PLDI 2021) alternates a wake phase (neurally-guided enumerative search) with sleep phases that refactor solutions into a growing library of reusable abstractions. LILO (Grand et al., ICLR 2024) replaces enumerative search with an LLM synthesiser, keeps symbolic compression via Stitch, and auto-documents abstractions so both humans and LLMs can reuse them. Top-down: VisProg (CVPR 2023 Best Paper) and ViperGPT (ICCV 2023) let an LLM write Python over off-the-shelf vision modules with no training at all. **That branch has essentially dissolved into LLM tool-use.**

Abstract reasoning is the hardest testbed. ARC (Chollet 2019) resists memorisation by construction. ARC-AGI-2 (2025) recurated it to resist brute-force program search: 1,000 training / 120 public eval / 120 semi-private / 120 private tasks, calibrated so every task was solved pass@2 by at least two humans. At release, pure LLMs scored ~0%.

One canonical ARC table (from the ARC Prize 2025 Technical Report, arXiv:2601.10904, Jan 2026 — 1,455 teams, 15,154 entries, 90 paper submissions):

System	ARC-AGI-2	Note
Top Kaggle entry (NVARC)	**24%** private eval	test-time training + Tiny-Recursive-Model components ⚠ *per-team attribution and $/task are single-source (arcprize.org blog)*
Frontier LLM w/ refinement harness	~54%	at roughly $30/task ⚠ *single-source; the two sources disagree on cost*
Tiny Recursive Model (7M params)	8%	45% on ARC-AGI-1; **won a paper prize at <0.01% of the parameters of the LLMs it beats**
Humans	not published as a single number	⚠ *no human-baseline percentage appears in either primary source — do not quote one*

The honest reading: **the winning entries are increasingly small transformers plus test-time training plus synthetic data, not anything recognisably symbolic.** Chollet's report names "the refinement loop" — per-task iterative program optimisation — as the defining methodological theme of 2025.

A completely different route to compositionality is vector-symbolic architectures / hyperdimensional computing: Smolensky's tensor products (1990), Plate's Holographic Reduced Representations (1995), Kanerva (2009), surveyed by Kleyko, Rachkovskij, Osipov & Rahimi in ACM CSUR. Binding and superposition as algebra on high-dimensional vectors, with resonator networks doing the un-binding. The flagship result is Hersche, Zeqiri, Benini, Sebastian & Rahimi, *A neuro-vector-symbolic architecture for solving Raven's progressive matrices* (Nature Machine Intelligence 2023): **87.7% on RAVEN, 88.1% on I-RAVEN**, with probabilistic reasoning two orders of magnitude faster than comparable symbolic search. Small, coherent, mostly neuromorphic-adjacent community (IBM Zurich, Redwood/Berkeley), real results, limited reach beyond RPM-shaped tasks.

3.6 Structure over graphs — knowledge graphs, rule learning, GNN expressiveness

Be blunt with yourself about what is and isn't neurosymbolic here.

The first wave — TransE (2013), DistMult (2015), ComplEx (2016), RotatE (2019) — learns one vector per entity and relation plus a fixed scoring function. Nothing is symbolic; the "reasoning" is bilinear algebra over a lookup table. What makes it *look* symbolic is that the algebraic form decides which relational patterns are representable at all: TransE's translation collapses symmetric relations to a zero vector; DistMult's diagonal bilinear form forces symmetry; ComplEx's Hermitian product buys antisymmetry; RotatE's complex rotation buys symmetry, antisymmetry, inversion and composition at once. Gutiérrez-Basulto & Schockaert (KR 2018) formalised the harder question — when can a vector-space representation be jointly consistent with a rule base — with the general answer that you need region/convex semantics, not points.

The purely symbolic line never died, and this is the most uncomfortable fact in the sub-area. AMIE+ mines closed Horn rules with open-world-aware confidence measures. AnyBURL (Meilicke, Chekol, Ruffinelli & Stuckenschmidt, IJCAI 2019; VLDB Journal 33(1):131–161, 2024) samples paths bottom-up and generalises them into rules, anytime — **CPU-only, minutes of training, competitive with heavily tuned embedding models on FB15k-237/WN18RR, and it emits human-readable rules.** That is either a strong argument for symbolic methods or a strong argument that the benchmarks reward pattern memorisation. Probably both.

The genuinely neurosymbolic entries are the ones whose *output artifact has logical semantics*: Neural-LP, DRUM, and RNNLogic (ICLR 2021), which treats rules as latent variables with an EM loop between a rule generator and a reasoning predictor.

The pivot: path-based GNNs. NBFNet (Zhu, Zhang, Xhonneux & Tang, NeurIPS 2021) neuralises the Bellman–Ford recursion — pair representations as a generalised sum over paths, each path a generalised product of edge representations — yielding *source-conditioned* node representations with no free entity embeddings. That makes it inductive over unseen entities and makes predictions decompose into paths, i.e. rule-like explanations. A*Net (NeurIPS 2023) adds a learned priority function visiting only ~10% of nodes/edges per iteration, reaching million-scale graphs. ULTRA (Galkin, Yuan, Mostafa, Tang & Zhu, ICLR 2024) removes the last vocabulary dependence by building relation representations from a graph of relation–relation interactions: **one pretrained model, zero-shot link prediction on 57 KGs, often on par with or better than per-graph-trained baselines.** Then Huang, Barceló, Bronstein, Ceylan, Galkin, Reutter & Romero Orth (ICML 2025) supply the theory: KGFM expressiveness is governed exactly by the *motifs* used to build relation representations, and the binary motifs everyone uses are a real ceiling.

The theory thread is the most honestly logical part. Barceló, Kostylev, Monet, Pérez, Reutter & Silva, *The Logical Expressiveness of Graph Neural Networks* (ICLR 2020), characterises which logical node classifiers messagepassing GNNs capture and shows a trivial architectural change (global readout) changes the logical class. Grohe's *The Descriptive Complexity of Graph Neural Networks* (TheoretiCS vol. 3, 2024) places GNN-computable queries in (uniform) TC^0 and a guarded fragment of first-order logic with counting. **Read these as bounds on what a GNN can *separate*, not as evidence that it deduces.**

Complex query answering (GQE → Query2Box → BetaE) makes logical operators geometric or probabilistic. CQD (ICLR 2021) then showed you can skip that entirely: keep a plain ComplEx link predictor, optimise continuously over the query DAG with t-norms, and gain substantially with far less training data. **This is the clearest case in the field of symbolic structure beating a learned black box.**

The 2024–26 mass of the field is LLM+KG, and almost none of it is neurosymbolic. Pan et al. (IEEE TKDE 2024) is the standard roadmap. GraphRAG (Edge et al., 2024) builds an LLM-extracted entity graph with community summaries; HippoRAG adds Personalized PageRank; LinkedIn's deployed KG-RAG reports +77.6% MRR and −28.6% median ticket resolution time. **In all of it the KG is an *index*, not a theory: no inference rule is applied, no consistency is checked, nothing is entailed.** RoG and Think-on-Graph are closer because they constrain generation to real KG paths. By 2026 the field is openly asking whether the graph index survives agentic search at all — see *Do We Still Need GraphRAG?* (arXiv:2604.09666), which finds agentic search narrows the gap on standard QA while GraphRAG retains an edge on multi-hop reasoning.

Two sub-areas usually missing from NeSy reviews:

- **Ontology / description-logic embeddings.** Kulmanov, Liu-Wei, Yan & Hoehndorf, *EL Embeddings* (2019), embed EL++ ontologies into $\mathbb{R}^n$ so concepts become n-balls and *the geometry is a model of the theory* — subsumption becomes containment, so the embedding is sound with respect to the logic rather than merely correlated with it. Evaluated on protein-protein interaction prediction. Successors: Box^2EL, TransBox, ALC saturation embeddings. This is simultaneously the semantic-web wing and the bioinformatics application.
- **Neural algorithmic reasoning.** Veličković & Blundell, *Neural Algorithmic Reasoning* (Patterns 2021), operationalised by the CLRS benchmark (ICML 2022, 30 classical algorithms): train a processor network to imitate an algorithm's step-by-step execution in latent space, then reuse it on noisy real inputs. **It answers the symbol-grounding bottleneck differently — don't extract symbols, extract the algorithm.** Active through 2025–26 (Discrete NAR, ICML 2025; Tropical Attention, NeurIPS 2025).

3.7 The LLM era — language models coupled to solvers and provers

This is where the field's centre of gravity now sits, and where its strongest evidence lives.

One architectural move, repeated: stop asking the language model to *do* the reasoning; ask it to *write down the problem* in a formal language a sound external engine can decide. The neural part becomes a semantic parser; the symbolic part becomes the oracle.

The 2023 cohort established the pattern across four formalisms

System	Formal target	Headline
Faithful CoT (Lyu et al.)	mixed	splits Translation / Problem Solving so the visible chain **is by construction** what produced the answer; beats CoT on 9 of 10 benchmarks
Logic-LM (Pan, Albalak, Wang & Wang, EMNLP Findings 2023)	FOL / CSP / SAT via Z3, Pyke, clingo	**+39.2% over standard prompting, +18.4% over CoT** across ProofWriter, PrOntoQA, FOLIO, LogicalDeduction, AR-LSAT; introduced the solver-error-message repair loop everyone copies
LINC (Olausson, Gu, Lipkin et al., EMNLP 2023, Outstanding Paper)	first-order logic → Prover9	**StarCoder+ at 15.5B beats GPT-4-with-CoT by 10 absolute points on ProofWriter** — the parser can be small if the prover is sound
SatLM (Ye, Chen, Dillig & Durrett, NeurIPS 2023)	declarative spec → SMT	a *declarative* spec sits closer to the problem text than a procedure does; **+23% over program-aided LMs** on a hard GSM subset
LLM + ASP (Yang, Ishay & Lee, Findings of ACL 2023)	answer set programming	reusable task-independent ASP modules; SOTA on bAbI, StepGame, CLUTRR, gSCAN

The recurring finding across all of them: once a formalisation is executable, the solver essentially never errs. Residual failures concentrate in *translation*. That single sentence is the most durable result in this sub-literature.

⚠ *Directionally well supported by LINC, Logic-LM and Ishay et al.; per-category error percentages were not independently verified, so don't cite a numeric breakdown.*

The program-aided line chose Python instead of logic

PAL (Gao et al.) and Program-of-Thoughts (Chen et al., TMLR 2023) offload arithmetic to an interpreter; ToRA (ICLR 2024) trained models on interleaved tool-call trajectories; Toolformer generalised to self-supervised API use. **By 2025–26 this quietly won: the code interpreter is the de facto symbolic layer of deployed systems, and "tool use" is the commercial name for LLM-plus-solver.**

Formal mathematics — the field's strongest empirical case

- **AlphaGeometry** (Trinh, Wu, Le, He & Luong, Nature 625(7995):476–482, 2024): a language model trained on ~100M synthetic theorems proposes auxiliary constructions guiding a symbolic deduction engine. **25 of 30 olympiad geometry problems vs 10 for the prior best**, no human proof data. AlphaGeometry 2 raised IMO-geometry coverage from 66% to 88%.
- **AlphaProof** (Hubert, Mehta, Sartran et al., Nature 651(8106):607–613, epub 12 Nov 2025): AlphaZero-style RL inside Lean over ~80M auto-formalised problems (~80,000 TPU-days for the main RL run), with test-time RL generating and training on variants of the target problem. **Solved 3 of the 5 non-geometry IMO 2024 problems, including P6 which only 5 human contestants solved; with AlphaGeometry 2, 28/42 — silver-medal standard, the first medal-level AI performance at the IMO.** This is the most complete neurosymbolic system in the literature: every output is machine-checked by the Lean kernel, so correctness does not depend on trusting the model.

Then the open-prover race, one of the fastest benchmark climbs in ML:

Date	System	miniF2F	PutnamBench
Aug 2024	DeepSeek-Prover-V1.5	63.5%	—
Feb 2025	Goedel-Prover	57.6% pass@32	7 problems
Apr 2025	Kimina-Prover (preview)	80.7% pass@8192	—
Apr 2025	DeepSeek-Prover-V2-671B	88.9%	49 / 658
Jul 2025	Seed-Prover	saturates miniF2F	>50%; **5/6 IMO 2025 problems formally proved in Lean**
Aug 2025	Goedel-Prover-V2-32B	88.1% / 90.4%	86 problems
Dec 2025	Seed-Prover 1.5	—	**88%**; 80% Fate-H, 33% Fate-X, 11/12 Putnam 2025 in 9 hours
Jun 2026	Goedel-Architect	**99.2% pass@1** (100% w/ NL seeding)	**75.6% pass@1** (88.8%, 597/672, w/ seeding); 4/6 IMO 2025, 3/6 USAMO 2026

Read this table with three caveats. (a) PutnamBench denominators differ across papers (640 theorems / 1,692 formalisations in the original; 658 and 672 in later work) — cross-paper percentages are not strictly comparable. (b) pass@1 vs pass@32 vs pass@8192 vs verifier-assisted pass@k are routinely conflated, and none of these numbers is compute-normalised. (c) **miniF2F is dead as a discriminator.** The bottleneck moved upstream to *autoformalization* — stating the theorem correctly, not proving it.

Are reasoning models neurosymbolic?

The pro argument: in RLVR (named in Tulu 3), the reward comes from a symbolic checker — a math answer-matcher, a compiler, a test suite — so **the verifier is the symbolic component**. DeepSeek-R1 (Nature 645(8081):633–638, 2025) showed reasoning behaviour emerges from pure RL against automatically checkable rewards, with no human-annotated traces.

The contra argument is empirical, and it is worth taking seriously:

- Turpin et al. (NeurIPS 2023) and Lanham et al. show CoT is often **not a faithful account** of the computation, and that larger models produce *less* faithful reasoning.
- GSM-Symbolic (Mirzadeh, Alizadeh, Shahrokhi, Tuzel, Bengio & Farajtabar, ICLR 2025) rebuilds GSM8K from symbolic templates: accuracy varies across instantiations of the same question and **drops by up to 65% when one logically inert clause is added.**
- Kambhampati, Valmeekam, Guan et al., *LLMs Can't Plan, But Can Help Planning in LLM-Modulo Frameworks* (ICML 2024) argues autoregressive models cannot self-verify at all.
- *The Illusion of Thinking* (Shojaee, Mirzadeh, Alizadeh, Horton, Bengio & Farajtabar, NeurIPS 2025) finds reasoning models collapse to zero accuracy past a complexity threshold while *reducing* their reasoning-token budget as problems get harder.
- **And the rebuttal**: Lawsen argues the collapse is an experimental artifact — Tower of Hanoi failures coincide with output token limits the models explicitly acknowledge, some River Crossing instances are provably unsolvable yet scored as failures, and asking for a generating function instead of an exhaustive move list restores high accuracy. Two further rebuttals appeared within a month.

The honest position: this argument rests on a definitional move, not on any demonstration that the model's internal computation is symbolic, and the empirical picture is methodologically fragile in both directions.

The Illusion-of-Thinking exchange is itself a good lesson in how brittle negative results about LLM reasoning are.

One thing that complicates the tidy narrative

At IMO 2025, Gemini Deep Think scored **35/42 (gold, 5 of 6 problems) end-to-end in natural language inside the 4.5-hour limit**, with no formalisation step. ⚠ *(Verified from DeepMind's blog, not peer review. A parallel OpenAI claim at 35/42 could not be verified during preparation.)* In the *same* window, Seed-Prover formally proved 5 of 6 of the same problems in Lean. **Both facts are true.** You will see the first cited as evidence that scale beat the hybrid and the second cited as evidence that the hybrid works; each of those is an editorial reading, not a finding. What is not in dispute: within one year, the same lab's general model matched its own hybrid's problem count *and removed the human formalisation step*.

3.8 RL, planning, and world models

The unifying idea: keep a neural network for perception and low-level control, but move the *temporal and logical structure of the task* into a symbolic object — an automaton, a temporal-logic formula, a program, or a PDDL domain — that can be inspected, decomposed, verified, or searched over. Four largely separate communities.

(1) Symbolic task specification. Reward machines (Toro Icarte, Klassen, Valenzano & McIlraith, ICML 2018; **JAIR 73:173–208, 2022** — read the journal version) replace the opaque scalar reward with a finite-state machine over a labelling function. Because the learner *sees* the automaton, it can do counterfactual off-policy updates in every automaton state at once, automated reward shaping, and task decomposition, with convergence guarantees in the tabular case. Expressive power = regular languages = LTLf/LDLf over finite traces. LTL2Action (ICML 2021) generalises over combinatorial task sets on the order of 10^{39} instances. Restraining Bolts (De Giacomo, Iocchi, Favorito & Patrizi, ICAPS 2019) is conceptually sharpest: the agent's features and the specifier's features are deliberately *different*, so an authority can constrain an agent it did not design.

(2) Shielding and safe RL — covered in §3.4. The recurring cost: shield synthesis needs a symbolic model of the dynamics and scales badly, which is why almost all evaluations are gridworlds, Pac-Man, or small safety suites.

(3) Program-structured and relational policies. PIRL (Verma, Murali, Singh, Kohli & Chaudhuri, ICML 2018) searches a DSL for a program policy using a trained DRL agent as an oracle — the "neural oracle guides symbolic search" recipe that recurs everywhere. VIPER distils a deep policy into a decision tree so it can be model-checked. Neural Logic Machines (ICLR 2019) and NLRL (Jiang & Luo, ICML 2019, built on dILP) make the *policy itself* relational so it generalises across object counts never seen in training — which flat DRL policies do not.

(4) Symbolic world models. Classical action-model acquisition (LOCM, FAMA) learns STRIPS/PDDL operators from traces. Since 2023 the dominant approach is LLM-assisted: Guan, Valmeekam, Sreedharan & Kambhampati (NeurIPS 2023) have GPT-4 draft PDDL for 40+ actions, repair it via validators and human feedback, then hand it to a sound planner. **This relocates the LLM from an unsound plan generator to a knowledge source whose output is checked by classical machinery, restoring soundness and completeness guarantees.**

The calibration device you should read. PlanBench (Valmeekam, Marquez, Olmo, Sreedharan & Kambhampati, NeurIPS 2023 D&B) builds a planning benchmark from IPC domains including *obfuscated* "Mystery" variants where predicate names are replaced by meaningless tokens — stripping away commonsense retrieval so only actual state-space search survives. Then Valmeekam, Stechly & Kambhampati (arXiv:2409.13373):

System	Blocksworld	Randomised Mystery Blocksworld	Cost
o1-preview	**97.8%** (587/600)	**37.3%** (224/600)	$42.12 per 100 instances
LLaMA 3.1 405B	62.6%	0.8%	—
Claude 3.5 Sonnet	54.8%	0%	—
Fast Downward (2000s-era planner)	**100%**	**100%**	0.265 s/instance, ~free

These are the most trustworthy quantitative claims in the whole sub-area, precisely because they are negative results with a classical baseline.

Where the symbolic boundary actually sits. MuZero and DreamerV3 learn latent dynamics and plan inside them, but the latents carry no compositional or relational commitment — no object identity, no operator preconditions, nothing a verifier can consume. That, not "learned vs given", is the line. The 2024–26 frontier is precisely about inventing predicates that cross it: VisualPredicator (ICLR 2025 Spotlight) introduces *neuro-symbolic predicates* whose truth values are computed by calling a VLM on the raw observation, and invents new ones online when planning fails; OneLife (ICLR 2026) learns dynamics as executable "programmatic laws" in stochastic environments.

Deployment reality. Reward machines, shields, PIRL, VIPER, NLM and NLRL live in gridworlds, Crafter clones, Atari, small MuJoCo suites and Pac-Man; no production deployment was found. What genuinely runs on hardware is the LLM-plus-skill-library branch — SayCan (CoRL 2022: 101 instructions, 84% plan / 74% execution success on a real office-kitchen robot), Code as Policies (ICRA 2023), and 2026 work like *Build on Priors* (arXiv:2604.03759), which constructs a PDDL domain and matching control policies from 1–30 unannotated demonstrations and runs on a real industrial forklift — and there the "symbolic" component is a Python program or a PDDL sketch, not anything verified.

---

4. State of the art, 2025–2026

The one-sentence summary: neuro-symbolic AI stopped being mainly a research programme about *architectures* and became mainly a research programme about *verifiers*.

arXiv papers matching "neurosymbolic": 39 (2023) → 75 (2024) → 143 (2025) → 115 in Jan–Jul 2026. ⚠ *Keyword match on one spelling only, so it undercounts absolute volume; use it as a trend, not a census. It is also single-source.* Real growth, decelerating, and the composition has shifted almost entirely from end-to-end differentiable logic to "LLM proposes, symbolic engine checks."

4.1 Formal mathematics: proof of concept, then saturation

Covered in §3.7. In roughly eighteen months the field went from AlphaProof's silver medal to open systems reporting 99.2% pass@1 on miniF2F. **The benchmark died faster than the problem was solved.** The bottleneck is now autoformalization: ITPEval reports 29.1% pass@1 for statement translation and 10.5% for proof translation, and miniF2F-Lean itself was audited and found flawed. ⚠ *These two 2026 arXiv IDs could not be independently re-verified; treat as directional.*

4.2 ARC as the public scoreboard

Covered in §3.5. ARC-AGI-3 (launched March 2026) moves to *interactive* environments with no stated goals, testing exploration, memory and self-set subgoals. Reported humans ~100%, frontier AI <1%. ⚠ *Single-source (arcprize.org).* Nothing in the current neurosymbolic toolkit — program executors, concept bottlenecks, VSAs — obviously transfers to that setting, and the refinement-loop trick that carried ARC-AGI-2 does not obviously apply to environments you must first explore.

4.3 The theory wing consolidated on one problem

Reasoning shortcuts is now the most *cumulative* sub-area in the field:

- van Krieken et al. (NeSy 2025) prove the independence assumption means a model **cannot represent uncertainty over the concept combinations where shortcuts live — so it cannot even detect that it is taking one.**
- Neurosymbolic Diffusion Models (NeurIPS 2025) is the constructive answer.
- Takemura et al. (KR 2026) formalise shortcut-freeness as a CSP and prove deciding it is **coNP-complete** (counting them #P-complete). ⚠ *arXiv:2604.23377 verified only via its abstract page; the KR 2026 acceptance is as stated there.*
- rsbench (Bortolotti, Marconato, Carraro, Morettin, van Krieken, Vergari, Teso & Passerini) is shared benchmark infrastructure — and includes *formal verification procedures that count the reasoning shortcuts a given task's constraints admit*.

A negative result with a decision procedure and a benchmark suite is what maturity looks like. If you want a research foothold in classical NeSy, this is where the ground is firmest.

4.4 The systems wing made circuits fast

KLay (ICLR 2025), Dolphin (ICML 2025), DeepLog, REASON (HPCA 2026). Note what all of these accelerate: *evaluating* an already-compiled circuit. **Compilation itself, and the size of the ground provenance formula, are still CPU-bound and can be exponential in the task.** That is the unattacked half.

DeepLog (Manhaeve, Colamonaco, Derkinderen, Adriaensen, Van Praet, De Raedt & Marra, IJCAI 2026 demo) is the field's admission that a dozen incompatible prototypes was the real adoption barrier: a PyTorch-native backend compiling multiple NeSy languages to optimised arithmetic circuits. If classical NeSy ever gets used by ordinary ML practitioners it will be through something like this.

4.5 How the community now defines itself

The RAIL principles (Chiatti, Cochez, Cornelio, Dumančić, d'Avila Garcez, Lamb, Morra, Niepert, Peharz, Speranzon, Stol, ten Teije, Thanapalasingam, van Harmelen, van Krieken, Vergari & Wang, Aug 2026) — a 17-author community position paper proposing **R**easoning, **A**ssurances, **I**nterfacing, **L**earning as a unified frame, deliberately broad enough to claim AlphaProof and AlphaEvolve.

Read it as both a reasonable reframing *and* a tacit admission that classical NeSy did not win on its own terms. Where a field draws its boundary tells you where it thinks its legitimacy comes from.

4.6 Deployment and funding signals

- **AWS Automated Reasoning checks** in Bedrock Guardrails went GA 6 Aug 2025, claiming "up to 99% verification accuracy" for policy-grounded hallucination detection. ⚠ *Vendor claim, no peer review, no named solver beyond "SMT-LIB syntax", no named customer in the case study.*

- **Manufacturing**: CausalTrace (AAAI-26 IAAI — a track that specifically vets *deployed* applications), CausalPulse at a Robert Bosch plant. ⚠ *Metrics from an arXiv search summary, not fetched abstracts.*
- **Discovery**: AlphaEvolve — LLM proposes and mutates programs, an automated symbolic evaluator selects them; found a 48-multiplication algorithm for 4×4 complex matrix multiplication, the first improvement over Strassen in 56 years, plus data-centre scheduling and circuit improvements. Its predecessor FunSearch (Nature 625:468–475, 2024) is the canonical LLM-plus-verifier discovery result. Georgiev, Gómez-Serrano, **Terence Tao** & Wagner then applied the loop to 67 open or semi-open problems (arXiv:2511.02864) — a working mathematician's audit of what neurosymbolic search actually contributes.
- **Funding**: DARPA **ANSR** (Assured Neuro Symbolic Learning and Reasoning, PM Susmit Jha, IPTO, BAA HR001122S0039) is the anchor US line. ⚠ *The programme page carries no 2025–26 status update, budget, or performer list.* No specific Horizon Europe project could be verified.

4.7 Institutions

- **NeSy conference** — 19th edition Sep 2025 (UC Santa Cruz, PMLR vol. 284); 20th edition Lisbon, 1–4 Sep 2026. The series began in 2005 as a workshop but has **not** run every year — "annual since 2005" would make 2025 the 21st. Run by the Neurosymbolic AI Association (nesy-ai.org).
- **IJCLR** — 5th edition Surrey Sep 2025, 6th UPV Valencia Sep 2026; leans inductive logic programming.
- **NeuS** (International Conference on Neuro-symbolic Systems) held a 2026 edition — usually missed in venue lists.
- ***Neurosymbolic Artificial Intelligence*** journal (IOS Press / SAGE, ISSN 2949-8732), open access. ⚠ *Sources disagree on the launch year; the journal homepage shows articles from 2024–25 onward. State it carefully.*
- **NeurIPS 2025 had no neurosymbolic-branded workshop** — the reasoning workshops are LLM-centric. That is itself a signal about where the attention went.

---

5. The honest scorecard

5.1 Read the flagship results critically

NS-VQA's 99.8% on CLEVR. The strong pure-neural baseline of that era, MAC (Hudson & Manning), was already at 98.9%; FiLM at 97.7%. **So the flagship NeSy vision-reasoning win was ~0.9 points, purchased with program annotations, on a synthetic benchmark that is now saturated.** This is the single best case study in how to read this literature.

MNIST-addition. See §3.2. Toy at N=2, unsolved at N=15.

SATNet's visual Sudoku. Depended on label leakage (Topan, Rolnick & Si, 2021).

5.2 The generalization claim has the most instructive arc

SCAN (2018), CFQ (2020) and gSCAN (2020) were built to show seq2seq models fail systematically, and structured/NeSy methods led for several years. Then neural-only methods largely dissolved them:

- **Least-to-most prompting** (Zhou et al., ICLR 2023) reached **≥99% on all SCAN splits including the length split using 14 exemplars**, versus 16% for chain-of-thought — while, in the authors' framing, "specialized neural-symbolic models required over 15,000 training examples."
- **Drozdov et al.** set CFQ state of the art at **95.0 mean over MCD1/2/3 using 1% of the training data**, beating LeAR's 90.9. ⚠ *Believed ICLR 2023; venue not confirmed.*

- **Lake & Baroni's MLC** (Nature 2023) matched human systematicity with a plain transformer.

The implication a newcomer should absorb: those benchmarks were measuring a training-distribution deficiency, not an irreducible architectural one, and the NeSy data-efficiency advantage largely evaporates once a large pretrained model is the baseline. Any data-efficiency argument made after 2022 has to answer this.

5.3 Reasoning shortcuts are pervasive — and not only in NeSy

rsbench on miniBOIA:

Model	Label accuracy	**Concept accuracy**
DeepProbLog	0.87	**0.28**
Logic Tensor Networks	0.78	**0.35**
CLIP	0.99	**0.34**

High task scores masking wrong concepts, in neural, neurosymbolic and foundation models alike. **The symbolic layer does not buy correct concepts.**

5.4 Documented methodological problems

- **Solver choice accounts for ~50% of reported variation** in LLM+solver pipelines (Lam, Thatikonda & Shareghi, arXiv:2406.00284), with an almost linear correlation between tool-executable rate and final accuracy. This is the clearest documented case of missing evaluation controls in modern NeSy.
- **The properties the field sells are the least worked on.** Colelough & Regli's PRISMA review (arXiv:2501.05435) screened 1,428 papers to 167 peer-reviewed, code-backed ones for 2020–2024: learning/inference 63%, knowledge representation 44%, logic/reasoning 35%, **explainability/trustworthiness 28%, metacognition 5%.**
- **No standard evaluation protocol.** Papers pick their own tasks, baselines and solvers. Nothing analogous to GLUE or ImageNet exists; rsbench is the closest candidate and is two years old.
- **The independence assumption is provably harmful** (§3.2) and near-universal.
- **Two careful audits reach different conclusions about which architectural family works best** — Hamilton et al. (Semantic Web journal, 2022) found logic-compiled-into-network systems satisfy the most stated NeSy *goals*; Chatzikyriakidis & Lappin (Frontiers in AI 9, article 1797587, 22 May 2026) report that "injective" systems gain 1–2% on general NLU while "federative" ones gain substantially (DSR-LM +20%, Logic-LM +39.2%). **Note carefully: these measure different quantities — goal satisfaction vs performance gain — so this is not strictly a disagreement.** But the uncomfortable observation stands: the field invests most effort in the tightly-coupled paradigms that deliver the least measured gain.

5.5 Where the evidence genuinely supports NeSy

Narrower and more specific than the marketing, but real:

1. **Hard constraint satisfaction by construction.** Semantic Probabilistic Layers give perfect constraint satisfaction. No amount of scale gives you that.
2. **Verification-in-the-loop formal reasoning.** AlphaGeometry's 25/30 vs 10 is a >2× jump on an externally defined problem set — the least-disputed win in the literature. AlphaProof's outputs are Lean-checked. Every accepted proof is *machine-verified*.
3. **Auditability and faithfulness by construction.** Faithful CoT and LINC buy faithfulness definitionally, for the fragment you can formalise.

4. **Semantic-level debuggability.** CLEVR-Hans: you can say "never use colour" to a symbolic scene representation; you cannot say it to a saliency map.
5. **Structured long-horizon control in low-data, energy-constrained regimes.** ⚠ *Suggested by a 2026 robotics result (95% vs 34% for a fine-tuned VLA baseline) but that is a single paper on Towers of Hanoi.*
6. **Constrained decoding.** Cheap, real, shipping everywhere.

Notice the pattern: every item on that list is symbolic *verification or search over neural proposals*, not a differentiable logic layer.

5.6 The verdict I would give a working data scientist

Treat "neurosymbolic" in 2026 as meaning, in practice, **"put a verifier in the loop."** That pattern is genuinely useful and now well evidenced. Treat the differentiable-logic literature as a separate, smaller, theoretically deeper research programme whose main current value to you is its *critique* — knowing why constraint satisfaction does not imply correct concept grounding transfers to any concept-based or constrained model you build.

And treat any claim that a differentiable logic layer will make your model more data-efficient or more compositional than a well-prompted frontier model as **unproven until you have run the baseline yourself.**

---

6. Open problems worth a researcher's time

Ranked by my read of tractability × importance, with the reason each is stuck.

1. **Reasoning-shortcut mitigation at realistic supervision budgets.** Detection is coNP-complete, counting #P-complete. Concept supervision works but is expensive. Bayesian-ensemble (BEARS) and prototype-based approaches are demonstrated only at rsbench scale. **Nothing scales to open-vocabulary perception.** This is the firmest ground in the field — negative results, a decision procedure, a benchmark — and therefore where incremental work compounds.
2. **A fair three-way comparison of ABL vs DeepProbLog vs LTN at matched supervision on a common suite.** Nobody has run it. Abduction avoids both #P-hard counting and fuzzy-gradient pathologies by solving a consistency problem instead; whether it actually scales better, and why, is unknown. This is an unusually cheap, unusually valuable contribution.
3. **Escaping conditional independence affordably.** It is what makes WMC factorise and what makes the model structurally unable to flag its own shortcut. Neurosymbolic diffusion is one answer; a general recipe for dependent symbol posteriors that keeps inference tractable does not exist.
4. **Autoformalization fidelity.** No scalable way to certify that a generated Lean/FOL/PDDL/SMT statement *means* the natural-language problem. A sound prover on a wrong formalisation gives a confidently wrong verified answer. This is now the binding constraint on the field's best sub-area.
5. **Compilation, not evaluation.** Every 2024–26 systems paper accelerates evaluating an already-compiled circuit. Incremental, amortised or reusable compilation across similar queries is an open gap.
6. **Symbol grounding under noise in RL.** Reward machines, shields and PDDL operators all assume a labelling function mapping raw states to ground-truth propositions; in practice that is a learned detector. Recasting it as a POMDP is known to be the right move and known not to scale. **This is the single biggest blocker to moving off gridworlds.**

7. **Grounding foundation models as probabilistic relations.** Vieira (Li et al., AAAI 2024, arXiv:2412.14515) makes an LLM callable as a *relation* inside probabilistic datalog — the bridge people claim doesn't exist. But LLM probabilities are uncalibrated and mutually dependent, violating exactly the conditional-independence assumption WMC requires. Whether calibrated, dependency-aware foundation-model predicates can be built at all is the actual obstacle.
8. **Distribution-faithful constrained decoding at production latency.** ASAp converges to the correct conditional distribution but needs repeated sampling; GUARD proves autoregressive training alone cannot reach it. **No method gives XGrammar-class latency AND distributional fidelity.**
9. **Combinatorics.** Neither AlphaProof nor AlphaGeometry 2 solved either IMO 2024 combinatorics problem (AlphaProof: 20.3% on formal-IMO combinatorics vs 75.7% number theory). LLMs aided by symbolic solvers also do poorly on first-order combinatorial problems. **This is the clearest remaining gap where a genuinely new hybrid could matter.**
10. **Ontology reasoning at scale with sound geometry.** EL++/ALC embeddings give models that satisfy axioms by construction, but there is no accepted evaluation separating deductive-closure recovery from statistical link prediction, and none handles the expressivity real biomedical ontologies use.
11. **Composition of guarantees.** A schema-valid tool call, a shielded action and a Lean-checked lemma each hold locally; nothing composes them into an end-to-end guarantee over a multi-step agent trajectory. Conformal monitoring gives marginal coverage, which does not compose across steps.
12. **A benchmark that separates reasoning from knowledge, and is neither saturated nor contaminated.** Measuring the thing NeSy claims to fix is itself unsolved.

---

7. A reading path

Sitting 1 — orientation (about 3 hours)

1. Garcez & Lamb, *Neurosymbolic AI: The 3rd Wave* — the field's own account and its vocabulary
2. Sarker, Zhou, Eberhart & Hitzler, *Neuro-symbolic artificial intelligence* (AI Communications 34(3), 2021) — the citable source for Kautz's six types
3. Van Harmelen & ten Teije, *A Boxology...* — so you see the taxonomy is contested

Sitting 2 — the technical map (one weekend)

4. Marra, Dumančić, Manhaeve & De Raedt, *From statistical relational to neurosymbolic AI* — the seven-dimension design space. **For a reader with your background this is the highest-value single paper.**
5. Badreddine et al., *Logic Tensor Networks* — the complete, teachable exposition of logic-as-loss, from t-norms to running code
6. Manhaeve et al., DeepProbLog — the neural predicate → WMC → differentiable circuit mechanism

Sitting 3 — the critique (read this before you believe sitting 2)

7. van Krieken, Acar & van Harmelen, *Analyzing Differentiable Fuzzy Logic Operators* — which of the operators you just learned actually produce usable gradients
8. Marconato et al., *Symbol Grounding in NeSy AI: A Gentle Introduction to Reasoning Shortcuts* — the 2026 synthesis of why constraint satisfaction ≠ concept correctness
9. Topan, Rolnick & Si, *Techniques for Symbol Grounding with SATNet* — the field's cleanest replication failure
10. van Krieken et al., A-NeSI — read *the baseline table*; it is the most honest scalability data in the field

Sitting 4 — the LLM era (where the field actually is)

11. Pan et al., Logic-LM — the cleanest statement of the paradigm
12. Olausson et al., LINC — the same idea done rigorously, with the error analysis showing translation (not proving) is the bottleneck
13. Hubert et al., AlphaProof (Nature 651:607–613) — the fully verified end state, and the compute bill
14. Yang et al., IJCAI 2025 survey — the Symbolic→LLM / LLM→Symbolic / LLM+Symbolic vocabulary

Sitting 5 — calibration

15. Valmeekam, Stechly & Kambhampati, *LLMs Still Can't Plan; Can LRMs?* — the 97.8%-vs-37.3%-vs-Fast-Downward's-100% table
16. Lake & Baroni, MLC (Nature 2023) — the strongest evidence *against* the field's central premise, which you should be able to argue with
17. Chollet et al., ARC Prize 2025 Technical Report — verified 2026 numbers with costs, so you can tell hype from result

Then pick one family from §3 and go deep. If you want the firmest ground: reasoning shortcuts (§3.1/§4.3). If you want the most impact: autoformalization (§3.7). If you want the least crowded: abductive learning (§3.3) or neural algorithmic reasoning (§3.6).

---

8. What you can actually run

Maintenance status checked 7–8 Aug 2026 via the GitHub API (last-push dates; these overstate liveness for repos whose only recent commits are dependency bumps).

Actively maintained — start here

Tool	What it is	Status
Scallop	Datalog-based NeSy language, Rust engine, PyTorch bindings, provenance semirings with a top-*k* dial	pushed Jun 2026, 502★, real docs at scallop-lang.org. **Try this first.**
ProbLog	The probabilistic logic programming substrate	Jun 2026, 416★
PyReason	Open-world temporal logic over knowledge graphs with explainable traces	Jul 2026, 345★
IBM LNN	Logical Neural Networks	Jul 2026, 325★
PyNeuraLogic	Lifted relational neural networks	Aug 2026, 312★
NeurASP	Neural networks inside answer set programming	Jul 2026, 54★
DomiKnowS	Declarative constraint integration	Aug 2026, 50★

The unbranded symbolic layer you probably already use — Z3 (12.5k★), clingo (817★), PySAT (460★), Lean/mathlib4 (3.8k★, daily commits), LeanDojo (824★, turns Lean into a Python-callable gym), SymPy, cvc5.

Constrained decoding — the most-deployed NeSy mechanism — outlines (15.5k★), XGrammar (1.8k★, default backend for vLLM/SGLang/TensorRT-LLM), llguidance (830★). All three pushed within days of Aug 2026.

Reference code only — do not build on these

DeepProbLog (last push Aug 2024, 351★ — effectively frozen; KU Leuven's effort moved to DeepLog), LTNtorch (Oct 2024, 41★), Pylon (Jan 2024, 122★ — abandoned), UCLA Semantic-Loss (2019, 63★ — abandoned), Logic-LM (Jun 2024, 404★ — baselines are GPT-3.5/GPT-4-era), Juice/ProbabilisticCircuits.jl (Jun 2024, 106★).

Benchmarks worth running — rsbench (concept quality + shortcut counting; run it before you believe any NeSy interpretability claim), MNIST-Addition (via DeepProbLog), CLEVR-Hans3/7, FOLIO (1,430 examples over 487 premise sets), ProofWriter, PrOntoQA, PlanBench/Blocksworld, ARC-AGI-2, miniF2F + PutnamBench (1,692 formalisations of 640 theorems), ZebraLogic, ROAD-R.

Teaching material — the `neurosym` library accompanying Chaudhuri, Ellis, Polozov, Singh, Solar-Lezama & Yue's *Neurosymbolic Programming* monograph (Foundations and Trends in Programming Languages, 2021 ⚠ *volume/pages unconfirmed — nowpublishers returns 403*); the ScaDS.AI School on Neuro+Symbolic AI (Leipzig, June 2026).

The honest signal in this table: the branded NeSy libraries are thinly maintained while the tools practitioners actually run every day are the unbranded symbolic layer. Treat the branded frameworks as research artifacts, with Scallop, ProbLog, PyReason, PyNeuraLogic and IBM's LNN as the genuine exceptions.

---

9. Positioning yourself

Three observations that follow from everything above.

The field's centre of gravity has left the field. The highest-profile neuro-symbolic results of 2024–26 — AlphaGeometry, AlphaProof, AlphaEvolve — came from DeepMind, not from NeSy regulars, and sit in Kautz's *loosest* categories. Meanwhile the tightly-coupled differentiable-logic research the founders cared about is small-scale and benchmark-bound. If you want impact, the LLM+verifier side is where it is. If you want depth and uncrowded ground, the classical side has firmer theory and a clearer notion of what would count as progress.

The best work in this field right now is self-critical. Reasoning shortcuts, the independence-assumption proof, rsbench, honest timeout tables, the SATNet replication — these came from inside the community, and they are the results that have held up. That is what a maturing subfield looks like. It also means the highest-leverage contribution a careful empiricist can make is often an evaluation, not a method.

Your comparative advantage as a data scientist is the baseline. The single most common defect in this literature is a NeSy method compared against a from-scratch neural model on a synthetic task. Where anyone has redone the comparison against a strong pretrained baseline — SCAN with 14 exemplars, CFQ with 1% of the data, MLC — the advantage disappeared. **Running the baseline properly is a publishable contribution in this field, which tells you something about the field.**

Full annotated bibliography, grouped by theme with verification status: Part II — Annotated Bibliography

Part II — Annotated Bibliography

Grouped by theme, roughly chronological within each group.

Verification convention. Every entry was checked against a bibliographic source (arXiv API, OpenAlex, dblp, or a publisher page). Entries carrying ⚠ have some element that could not be confirmed — the note says which. Entries marked **[corrected]** had a metadata error in circulation that has been fixed here; the correction is stated. Roughly 75 citations were independently adjudicated in a dedicated audit pass: **zero fabrications, zero invented first authors, zero invented venues** were found; 12 metadata errors were.

---

1. Foundations and history

- **McCulloch, W.S. & Pitts, W.** (1943). *A logical calculus of the ideas immanent in nervous activity.* Bulletin of Mathematical Biophysics 5:115–133. link — Neurons as threshold logic units; networks realise arbitrary propositional logic. The first neural network was a logic machine, which is why NeSy frames itself as reunification.
- **Fodor, J.A. & Pylyshyn, Z.W.** (1988). *Connectionism and cognitive architecture: A critical analysis.* Cognition 28(1–2):3–71. link — The systematicity challenge. Every compositional-generalization benchmark descends from this argument.
- **Smolensky, P.** (1988). *On the proper treatment of connectionism.* Behavioral and Brain Sciences 11:1–23.
- **Smolensky, P.** (1990). *Tensor Product Variable Binding and the Representation of Symbolic Structures in Connectionist Systems.* Artificial Intelligence 46(1–2):159–216. dblp — Exact, invertible variable binding inside a vector space. Ancestor of all vector-symbolic architectures. *[corrected]* Title ends "Connectionist **Systems**", not "Networks" (several citation lists have this wrong).
- **Towell, G.G. & Shavlik, J.W.** (1994). *Knowledge-Based Artificial Neural Networks (KBANN).* Artificial Intelligence 70(1–2):119–165. dblp — Compile a propositional domain theory into network topology and weights, then refine by backprop. The template for all knowledge injection.
- **Valiant, L.G.** (2000). *Robust logics.* Artificial Intelligence 117(2):231–253. DOI 10.1016/S0004-3702(00)00002-3. (STOC 1999 predecessor at pp. 642–651.)
- **Valiant, L.G.** (2003). *Three problems in computer science.* JACM 50(1):96–99. link — Names reconciling statistical learning with logical reasoning as a grand challenge. The field's most-cited legitimacy argument.
- **d'Avila Garcez, A., Lamb, L.C. & Gabbay, D.M.** (2009). *Neural-Symbolic Cognitive Reasoning.* Springer, Cognitive Technologies series. — Neural encodings of modal, temporal, epistemic and intuitionistic logics with a general translation-and-extraction methodology.
- **Sun, R.** (2024). *Dual-process theories, cognitive architectures, and hybrid neural-symbolic models.* Neurosymbolic Artificial Intelligence vol. 1. DOI 10.3233/NAI-240720. *[corrected]* OpenAlex dates this **2024**, not 2025. — Argues modern System-1/System-2 enthusiasm rediscovers 1990s CLARION-era principles. ⚠ Sun's CONSYDERR book (Wiley 1994) could not be verified and is omitted.

---

2. Position papers, taxonomies and surveys

- **Marcus, G.** (2018). *Deep Learning: A Critical Appraisal.* arXiv:1801.00631
- **Marcus, G.** (2020). *The Next Decade in AI: Four Steps Towards Robust Artificial Intelligence.* arXiv:2002.06177

- **Chollet, F.** (2019). *On the Measure of Intelligence.* arXiv:1911.01547 — Intelligence as skill-acquisition efficiency relative to priors; introduces ARC.
- **van Harmelen, F. & ten Teije, A.** (2019). *A Boxology of Design Patterns for Hybrid Learning and Reasoning Systems.* Journal of Web Engineering 18(1–3):97–124. arXiv:1905.12389 — The principal competitor to Kautz's taxonomy; compositional where Kautz is a single ordinal scale.
- **d'Avila Garcez, A. & Lamb, L.C.** (2020). *Neurosymbolic AI: The 3rd Wave.* arXiv:2012.05876; Artificial Intelligence Review 2023, DOI 10.1007/s10462-023-10448-w. — The field's self-definition document.
- **Sarker, M.K., Zhou, L., Eberhart, A. & Hitzler, P.** (2021). *Neuro-symbolic artificial intelligence.* AI Communications 34(3):197–209. DOI *[corrected]* The published journal title is *Neuro-symbolic artificial intelligence*; "…: Current Trends" is the arXiv:2105.05330 **preprint** title only. Issue number is 3. **This is the safe citation for the Kautz taxonomy.**
- **Kautz, H.A.** (2022). *The Third AI Summer: AAAI Robert S. Engelmore Memorial Lecture.* AI Magazine 43(1):93–104. DOI 10.1609/aimag.v43i1.19122 *[corrected]* dblp gives 43(1):**93–104**; the widely-circulated "43(1):105–125 / DOI 10.1002/aaai.12036" variant has the wrong page range. ⚠ **The six-type taxonomy could not be verified against the printed text** (Wiley returns HTTP 402; AAAI OJS errors). Cite Sarker et al. 2021 for the taxonomy itself.
- **Hamilton, K., Nayak, A., Božić, B. & Longo, L.** (2022). *Is Neuro-Symbolic AI Meeting its Promise in Natural Language Processing? A Structured Review.* Semantic Web journal, DOI 10.3233/SW-223228. arXiv:2202.12205 — Audits five claimed NeSy goals against published NLP systems.
- **Giunchiglia, E., Stoian, M.C. & Lukasiewicz, T.** (2022). *Deep Learning with Logical Constraints.* IJCAI 2022 survey track, pp. 5478–5485. arXiv:2205.00523 — The best map of the constraint terrain; makes the soft/hard axis explicit.
- **Goyal, A. & Bengio, Y.** (2022). *Inductive biases for deep learning of higher-level cognition.* Proc. R. Soc. A 478(2266):20210068. arXiv:2011.15091 — The citable proxy for the System-1/System-2 framing. ⚠ Bengio's NeurIPS 2019 keynote has no citable published record.
- **LeCun, Y.** (2022). *A Path Towards Autonomous Machine Intelligence,* v0.9.2. OpenReview
- **Hitzler, P., Sarker, M.K. & Eberhart, A.** (eds.) (2023). *Compendium of Neurosymbolic Artificial Intelligence.* IOS Press, Frontiers in AI and Applications vol. 369. *[corrected]* The publisher page shows **24 chapters** running to page 546 — the "704 pp., 30 chapters" figure in circulation is unsupported. Predecessor: *Neuro-Symbolic Artificial Intelligence: The State of the Art,* FAIA vol. 342 ⚠ (publisher says 2021, commonly cited as 2022 — unresolved).
- **Chaudhuri, S., Ellis, K., Polozov, O., Singh, R., Solar-Lezama, A. & Yue, Y.** (2021). *Neurosymbolic Programming.* Foundations and Trends in Programming Languages 7(3):158–243. ⚠ Volume/issue/pages unverified (nowpublishers returns 403; not indexed under that title in OpenAlex). The monograph is real; the locator is not confirmed. Teaching library: `pip install neurosym`.
- **Marra, G., Dumančić, S., Manhaeve, R. & De Raedt, L.** (2024). *From statistical relational to neurosymbolic artificial intelligence: A survey.* Artificial Intelligence 328:104062. arXiv:2108.11451 — **The best technical map for an ML reader.** Seven design dimensions. Every volume/page detail verified.
- **Wang, W., Yang, Y. & Wu, F.** (2024). *Towards Data- and Knowledge-Driven Artificial Intelligence: A Survey on Neuro-Symbolic Computing.* IEEE TPAMI. arXiv:2210.15889 — The highest-visibility non-Western synthesis; partitions the field differently.
- **DeLong, L.N., Fernández Mir, R. & Fleuriot, J.D.** (2025). *Neurosymbolic AI for Reasoning over Knowledge Graphs: A Survey.* IEEE TNNLS 36(5):7822–7842. arXiv:2302.07200 — Useful precisely because it forces the neurosymbolic-vs-graph-ML distinction.

- **Colelough, B.C. & Regli, W.** (2025). *Neuro-Symbolic AI in 2024: A Systematic Review.* arXiv:2501.05435 — PRISMA review, 1,428 papers screened to 167 peer-reviewed and code-backed. Learning/inference 63%, KR 44%, logic/reasoning 35%, explainability/trust 28%, metacognition 5%.
- **Velasquez, A., Bhatt, N., Topcu, U., Wang, Z., Sycara, K., Stepputtis, S., Neema, S. & Vallabha, G.** (2025). *Neurosymbolic AI as an antithesis to scaling laws.* PNAS Nexus 4(5):pgaf117. link *[corrected]* Second author is **Neel** Bhatt, not "Nikhil". Efficiency claims traced: "100× smaller than GPT-3" → West et al. 2022; "0.1% training time / 1% data / 96.9% fewer parameters" → Zhao et al. 2024 (physics-informed autonomous driving) — domain-specific, not general.
- **Yang, X.-W., Shao, J.-J., Guo, L.-Z., Zhang, B.-W., Zhou, Z., Jia, L.-H., Dai, W.-Z. & Li, Y.-F.** (2025). *Neuro-Symbolic Artificial Intelligence: Towards Improving the Reasoning Abilities of Large Language Models.* IJCAI 2025 Survey Track, pp. 10770–10778. arXiv:2508.13678 — The Symbolic→LLM / LLM→Symbolic / LLM+Symbolic taxonomy. Companion repo: LAMDA-NeSy/Awesome-LLM-Reasoning-with-NeSy.
- **De Smet, L. & De Raedt, L.** (2025). *Defining neurosymbolic AI.* arXiv:2507.11127
- **Chatzikyriakidis, S. & Lappin, S.** (2026). *Neuro-symbolic NLP: taxonomy, assessment, and directions.* Frontiers in Artificial Intelligence, vol. 9, article 1797587, published 22 May 2026. DOI — Injective vs federative × five interaction modes. Reports injective systems gaining 1–2% on general NLU vs substantial gains for federative ones.
- **Chiatti, A., Cochez, M., Cornelio, C., Dumančić, S., d'Avila Garcez, A., Lamb, L.C., Morra, L., Niepert, M., Peharz, R., Speranzon, A., Stol, M., ten Teije, A., Thanapalasingam, T., van Harmelen, F., van Krieken, E., Vergari, A. & Wang, B.** (2026). *The RAIL Principles for Neurosymbolic AI.* arXiv:2608.04285 — 17-author community position paper; deliberately broad enough to claim AlphaProof and AlphaEvolve.

---

3. Fuzzy / differentiable logic (logic as a loss or a layer)

- **Diligenti, M., Gori, M. & Saccà, C.** (2015 online / 2017 print). *Semantic-based regularization for learning and inference.* Artificial Intelligence 244:143–165. DOI *[corrected]* Volume and pages confirmed; the DOI stamps it **2015 online-first** — "2017" is the print-issue year.
- **Rocktäschel, T. & Riedel, S.** (2017). *End-to-End Differentiable Proving.* NIPS 2017. arXiv:1705.11040 — Unrolls Prolog backward chaining, replaces unification with an RBF kernel on symbol embeddings. Beautiful; the proof tree explodes.
- **Yang, F., Yang, Z. & Cohen, W.W.** (2017). *Differentiable Learning of Logical Rules for Knowledge Base Reasoning (Neural LP).* NIPS 2017. arXiv:1702.08367
- **Evans, R. & Grefenstette, E.** (2018). *Learning Explanatory Rules from Noisy Data (dILP).* JAIR 61:1–64, DOI 10.1613/jair.5714; IJCAI 2018 extended abstract pp. 5598–5602. Volume and pages verified.
- **Xu, J., Zhang, Z., Friedman, T., Liang, Y. & Van den Broeck, G.** (2018). *A Semantic Loss Function for Deep Learning with Symbolic Knowledge.* ICML 2018, PMLR 80. link — The cheapest on-ramp: one extra loss term.
- **Fischer, M., Balunović, M., Drachsler-Cohen, D., Gehr, T., Zhang, C. & Vechev, M.** (2019). *DL2: Training and Querying Neural Networks with Logic.* ICML 2019. PDF — Non-fuzzy translation; loss exactly zero when satisfied. ⚠ PMLR page not fetched.
- **Marra, G., Giannini, F., Diligenti, M. & Gori, M.** (2019). *LYRICS: a General Interface Layer to Integrate Logic Inference and Deep Learning.* ECML-PKDD 2019. arXiv:1903.07534 ⚠ Pagination unconfirmed; 2019 (conference) vs 2020 (proceedings print) ambiguous.

- **Riegel, R., Gray, A., Luus, F., Khan, N., Makondo, N., Akhalwaya, I., Qian, H., Fagin, R., Barahona, F., Sharma, U., Ikbal, S., Karanam, H., Neelam, S., Likhyani, S. & Srivastava, S.** (2020). *Logical Neural Networks.* arXiv:2006.13155 — Full 15-author list verified. The arXiv comment reads "In submission to NeurIPS 2020"; it was **never formally published there** — cite as arXiv 2020.
- **van Krieken, E., Acar, E. & van Harmelen, F.** (2022). *Analyzing Differentiable Fuzzy Logic Operators.* Artificial Intelligence 302:103602. arXiv:2002.06100 — **The single most important negative result in the sub-area.** Most canonical operators are unusable; the best-performing combinations violate ordinary logical laws.
- **Serafini, L. & d'Avila Garcez, A.** (2016). *Logic Tensor Networks: Deep Learning and Logical Reasoning from Data and Knowledge.* arXiv:1606.04422 — Verified; restore this to reading lists.
- **Badreddine, S., d'Avila Garcez, A., Serafini, L. & Spranger, M.** (2022). *Logic Tensor Networks.* Artificial Intelligence 303:103649. arXiv:2012.13635 — 68 pages, worked examples across seven task types. **The definitive teachable reference.** PyTorch port: LTNtorch (Carraro, Serafini & Aiolli, arXiv:2409.16045) ⚠ *[corrected]* author order is Carraro, **Serafini, Aiolli**; the claimed JMLR/MLOSS publication is unverified.
- **Petersen, F., Borgelt, C., Kuehne, H. & Deussen, O.** (2022). *Deep Differentiable Logic Gate Networks.* NeurIPS 2022. arXiv:2210.08277 — >1M MNIST images/sec on one CPU core.
- **Ciravegna, G., Barbiero, P., Giannini, F., Gori, M., Lió, P., Maggini, M. & Melacci, S.** (2022). *Logic Explained Networks.* Artificial Intelligence, article 103822, DOI 10.1016/j.artint.2022.103822. arXiv:2108.05149 *[corrected]* Year is **2022**, not 2023; full seven-author list confirmed verbatim.
- **Petersen, F., Kuehne, H., Borgelt, C., Welzel, J. & Ermon, S.** (2024). *Convolutional Differentiable Logic Gate Networks.* NeurIPS 2024 (Oral). proceedings — **86.29% CIFAR-10, 61M logic gates, 29× smaller than prior SOTA** (all verbatim from the abstract). *[corrected]* The "4 nanosecond inference" and "29×–61×" figures are **not in the paper** — they came from a social-media post. Drop them.
- **Yousefi, S., Plesner, A., Aczel, T. & Wattenhofer, R.** (2025). *Mind the Gap: Removing the Discretization Gap in Differentiable Logic Gate Networks.* **NeurIPS 2025 (main track)** — venue upgraded during audit. arXiv:2506.07500 ⚠ The "4.5× faster training / 98% gap reduction" figures remain unverified.
- **Ślusarz, N., Komendantskaya, E., Daggitt, M., Stewart, R. & Stark, K.** (2023). *Logic of Differentiable Logics.* LPAR 2023. arXiv:2303.10650 — Supplies formal semantics for differentiable logics and targets *provable specification satisfaction* rather than accuracy — a different objective most reviews of this area never consider.
- **van Krieken, E.** (2024). *Optimisation in Neurosymbolic Learning Systems.* PhD dissertation. arXiv:2401.10819 — Verified; safe to cite.

Rule learning (the differentiable-ILP branch)

- **Minervini, P., Riedel, S., Stenetorp, P., Grefenstette, E. & Rocktäschel, T.** (2020). *Learning Reasoning Strategies in End-to-End Differentiable Proving.* ICML 2020. arXiv:2007.06477 *[corrected]* **This is the paper title.** "Conditional Theorem Provers" is the *method* name and must not be used as the title — a very common miscitation. ⚠ PMLR pagination unverified.
- **Sadeghian, A., Armandpour, M., Ding, P. & Wang, D.Z.** (2019). *DRUM: End-To-End Differentiable Rule Mining on Knowledge Graphs.* arXiv:1911.00055 ⚠ NeurIPS 2019 venue from background, not from the arXiv page.
- **Qu, M., Chen, J., Xhonneux, L.-P., Bengio, Y. & Tang, J.** (2021). *RNNLogic: Learning Logic Rules for Reasoning on Knowledge Graphs.* ICLR 2021. arXiv:2010.04029

---

4. Probabilistic NeSy — weighted model counting, knowledge compilation,

circuits

- **Darwiche, A. & Marquis, P.** (2002). *A Knowledge Compilation Map.* JAIR 17:229–264. arXiv:1106.1819 — Decomposability + determinism ⇒ linear-time model counting. **The theory that makes the whole sub-area possible.**
- **Richardson, M. & Domingos, P.** (2006). *Markov logic networks.* Machine Learning 62(1–2):107–136. link — The instructive failure case: undirected, so no compact differentiable circuit.
- **De Raedt, L., Kimmig, A. & Toivonen, H.** (2007). *ProbLog: A Probabilistic Prolog and its Application in Link Discovery.* IJCAI 2007, pp. 2462–2467. — The distribution semantics everything else varies on.
- **Poon, H. & Domingos, P.** (2011). *Sum-Product Networks: A New Deep Architecture.* UAI 2011. ⚠ Not fetched.
- **Kisa, D., Van den Broeck, G., Choi, A. & Darwiche, A.** (2014). *Probabilistic Sentential Decision Diagrams.* KR 2014. ⚠ Not fetched.
- **Fierens, D., Van den Broeck, G., Renkens, J., Shterionov, D., Gutmann, B., Thon, I., Janssens, G. & De Raedt, L.** (2015). *Inference and learning in probabilistic logic programs using weighted Boolean formulas.* TPLP 15(3):358–401. DOI — Confirmed exactly as cited. **The single most load-bearing citation in this thread.**
- **Manhaeve, R., Dumančić, S., Kimmig, A., Demeester, T. & De Raedt, L.** (2018). *DeepProbLog: Neural Probabilistic Logic Programming.* NeurIPS 2018, pp. 3753–3763. proceedings Journal version: *Neural probabilistic logic programming in DeepProbLog,* Artificial Intelligence vol. 298, **article 103504** (2021), DOI. *[corrected]* The locator is article number **103504**, not "pages 103–504" — confirmed mangling. Origin of MNIST-addition.
- **Yang, Z., Ishay, A. & Lee, J.** (2020). *NeurASP: Embracing Neural Networks into Answer Set Programming.* IJCAI 2020. PDF
- **Huang, J., Li, Z., Chen, B., Samel, K., Naik, M., Song, L. & Si, X.** (2021). *Scallop: From Probabilistic Deductive Databases to Scalable Differentiable Reasoning.* NeurIPS 2021, pp. 25134–25145. — Top-*k*-proofs provenance. VQA margins of **+12.42%** over neural-module-network and **+21.66%** over transformer baselines, both verified verbatim.
- **Ahmed, K., Teso, S., Chang, K.-W., Van den Broeck, G. & Vergari, A.** (2022). *Semantic Probabilistic Layers for Neuro-Symbolic Learning.* NeurIPS 2022. arXiv:2206.00426 — Constraint circuit × probabilistic circuit, exactly renormalised. **The reference guarantee layer.**
- **Winters, T., Marra, G., Manhaeve, R. & De Raedt, L.** (2022). *DeepStochLog: Neural Stochastic Logic Programming.* AAAI 2022 36(9):10090–10100.
- **Aspis, Y., Broda, K., Lobo, J. & Russo, A.** (2022). *Embed2Sym.* KR 2022.
- **Li, Z., Huang, J. & Naik, M.** (2023). *Scallop: A Language for Neurosymbolic Programming.* PACMPL 7(PLDI):**1463–1487**, DOI 10.1145/3591280. *[corrected]* Pages per OpenAlex; the frequently cited "Article 166" is unverified. ⚠ Lead author is **Ziyang Li** (UPenn) — commonly misattributed.
- **Pryor, C., Dickens, C., Augustine, E., Albalak, A., Wang, W. & Getoor, L.** (2023). *NeuPSL: Neural Probabilistic Soft Logic.* IJCAI 2023. arXiv:2205.14268 ⚠ Claims from abstract summary only.
- **van Krieken, E., Thanapalasingam, T., Tomczak, J.M., van Harmelen, F. & ten Teije, A.** (2023). *A-NeSI: A Scalable Approximate Method for Probabilistic Neurosymbolic Inference.* NeurIPS 2023. arXiv:2212.12393 — Learned surrogates for the counting step. **Its baseline table is the most honest scalability data in the field:** DeepProbLog 97.20% (N=1), 95.20% (N=2), timeout (N=4); A-NeSI 75.90 ± 2.21 at N=15.
- **van Krieken, E., Minervini, P., Ponti, E.M. & Vergari, A.** (2024). *On the Independence Assumption in Neurosymbolic Learning.* ICML 2024, PMLR 235:49078–49097. arXiv:2404.08458

- **Maene, J., Derkinderen, V. & De Raedt, L.** (2024). *On the Hardness of Probabilistic Neurosymbolic Learning.* arXiv:2406.04472 — Introduces WeightME, an unbiased gradient estimator using a logarithmic number of SAT calls. ⚠ Suspected ICML 2024; venue unconfirmed.
- **Liu, A., Ahmed, K. & Van den Broeck, G.** (2024). *Scaling Tractable Probabilistic Circuits: A Systems Perspective* (PyJuice). arXiv:2406.00766 — Authors confirmed. ⚠ Venue and speedup figures unverified.
- **Maene, J., Derkinderen, V. & Zuidberg Dos Martires, P.** (2025). *KLay: Accelerating Arithmetic Circuits for Neurosymbolic AI.* ICLR 2025. arXiv:2410.11415
- **Naik, A., Liu, J., Wang, C., Sethi, A., Dutta, S., Naik, M. & Wong, E.** (2025). *Dolphin: A Programmable Framework for Scalable Neurosymbolic Learning.* ICML 2025, PMLR 267:45461–45483. arXiv:2410.03348 — **1.71×–62× speedups over 13 benchmarks**; converges where Scallop, ISED and IndeCateR+ do not.
- **van Krieken, E., Minervini, P., Ponti, E.M. & Vergari, A.** (2025). *Neurosymbolic Diffusion Models.* NeurIPS 2025. arXiv:2505.13138
- **Derkinderen, V., Manhaeve, R., Adriaensen, R., Van Praet, L., De Smet, L., Marra, G. & De Raedt, L.** (2025/26). *The DeepLog Neurosymbolic Machine.* arXiv:2508.13697; demo paper *DeepLog: A Software Framework for Modular Neurosymbolic AI,* IJCAI 2026, arXiv:2605.10279.
- **Jiao, Y., Castellano Ontiveros, R., De Raedt, L., Gori, M., Giannini, F., Diligenti, M. & Marra, G.** (2026). *DeepProofLog: Efficient Proving in Deep Stochastic Logic Programs.* AAAI 2026 (Oral). arXiv:2511.08581 — Recasts resolution as an MDP so DP/RL replace explicit provenance construction.
- **Wan, Z., Liu, C.-K., Qian, J., Yang, H., Raychowdhury, A. & Krishna, T.** (2026). *REASON: Accelerating Probabilistic Logical Reasoning…* IEEE HPCA 2026. arXiv:2601.20784 — 12–50× over GPUs, 310–681× energy efficiency, 6 mm^2 at 2.12 W on TSMC 28 nm.
- **Li, Z., Huang, J., Liu, J., Zhu, F., Zhao, E., Dodds, W., Velingker, N., Alur, R. & Naik, M.** (2024). *Relational Programming with Foundation Models* (Vieira). AAAI 2024, DOI 10.1609/aaai.v38i9.28934. arXiv:2412.14515 — **The actual bridge between probabilistic NeSy programming and foundation models** that most reviews claim does not exist: an LLM invoked as a *relation* inside a probabilistic logic program, its outputs flowing through the provenance semiring.

5. Abductive learning and differentiable ASP

- **Dai, W.-Z., Xu, Q., Yu, Y. & Zhou, Z.-H.** (2019). *Bridging Machine Learning and Logical Reasoning by Abductive Learning.* NeurIPS 2019. proceedings — Perception proposes pseudo-labels, a symbolic KB abduces the most consistent revision, retrain, repeat. **Non-differentiable, consistency-driven — an architectural family absent from most Western surveys.** Predecessor: arXiv:1802.01173 (2018).
- **Hu, et al.** (2025). AAAI 2025 Oral. arXiv:2412.08457
- **Hu, et al.** (2025). *Curriculum Abductive Learning.* NeurIPS 2025. arXiv:2505.12275
- **Gao, Inoue, et al.** (2025). *Differentiable Rule Induction from Raw Sequence Inputs.* ICLR 2025.
- **Baugh, Perreault, Baugh, Dickens, Inoue & Russo** (2025). *Disentangling Neural Disjunctive Normal Form Models.* NeSy 2025. arXiv:2507.10546
- **Eiter, T., Inoue, K. & Moriyama, S.** (2026). *Neural Decision-Propagation for Answer Set Programming.* IJCAI-ECAI 2026. arXiv:2605.01797
- **Takemura, A. & Inoue, K.** (2026). *Differentiable Logic Programming to Mitigate Reasoning Shortcuts in Neurosymbolic Systems.* ICLP 2026, EPTCS 450:29–51. arXiv:2607.21185

- **Rader & Russo** (2026). *Accelerating NeurASP with vectorization and caching.* TPLP (ICLP 2026). arXiv:2606.10787

---

6. Hard constraints, guarantees, verification

- **Katz, G., Barrett, C., Dill, D., Julian, K. & Kochenderfer, M.** (2017). *Reluplex: An Efficient SMT Solver for Verifying Deep Neural Networks.* CAV 2017. arXiv:1702.01135 Successor: Marabou 2.0, CAV 2024 ⚠ (arXiv ID 2401.14461 from an ADS listing, not fetched).
- **Alshiekh, M., Bloem, R., Ehlers, R., Könighofer, B., Niekum, S. & Topcu, U.** (2018). *Safe Reinforcement Learning via Shielding.* AAAI 2018, pp. 2669–2678. arXiv:1708.08611
- **Wang, P.-W., Donti, P.L., Wilder, B. & Kolter, Z.** (2019). *SATNet: Bridging deep learning and logical reasoning using a differentiable satisfiability solver.* ICML 2019. arXiv:1905.12149 — **Learn the constraints** rather than inject them. Read with the replication failure below.
- **Topan, S., Rolnick, D. & Si, X.** (2021). *Techniques for Symbol Grounding with SATNet.* arXiv:2106.11072 — Showed the headline visual-Sudoku result depended on **label leakage. The field's cleanest replication failure, and it predates the reasoning-shortcut literature by two years.** ⚠ NeurIPS 2021 venue unconfirmed.
- **Vlastelica, M., Paulus, A., Musil, V., Martius, G. & Rolínek, M.** (2020). *Differentiation of Blackbox Combinatorial Solvers.* ICLR 2020 (spotlight). arXiv:1912.02175 — Informative gradients for exact solvers, one extra solver call per backward pass. Source of the Warcraft-shortest-path benchmark.
- **Giunchiglia, E. & Lukasiewicz, T.** (2020). *Coherent Hierarchical Multi-Label Classification Networks* (C-HMCNN). NeurIPS 2020.
- **Donti, P.L., Rolnick, D. & Kolter, J.Z.** (2021). *DC3: A learning method for optimization with hard constraints.* ICLR 2021. arXiv:2104.12225 ⚠ The "78× faster than qpth" figure could not be confirmed on the arXiv abstract.
- **Hoernle, N., Karampatsis, R.M., Belle, V. & Gal, K.** (2022). *MultiplexNet: Towards Fully Satisfied Logical Constraints in Neural Networks.* AAAI 2022. arXiv:2111.01564
- **Geng, S., Josifoski, M., Peyrard, M. & West, R.** (2023). *Grammar-Constrained Decoding for Structured NLP Tasks without Finetuning.* EMNLP 2023, pp. 10932–10952. ACL Anthology
- **Willard, B.T. & Louf, R.** (2023). *Efficient Guided Generation for Large Language Models* (Outlines). arXiv:2307.09702 ⚠ No peer-reviewed venue found.
- **Yang, W.-C., Marra, G., Rens, G. & De Raedt, L.** (2023). *Safe Reinforcement Learning via Probabilistic Logic Shields.* IJCAI 2023, pp. 5739–5749 (distinguished paper). arXiv:2303.03226
- **Park, K., Wang, J., Berg-Kirkpatrick, T., Polikarpova, N. & D'Antoni, L.** (2024). *Grammar-Aligned Decoding.* NeurIPS 2024. arXiv:2405.21047 — **The single most important gotcha paper for anyone shipping constrained decoding.** Masking is greedy local filtering, not Bayesian conditioning.
- **Dong, Y., Ruan, C.F., Cai, Y., Lai, R., Xu, Z., Zhao, Y. & Chen, T.** (2024/25). *XGrammar: Flexible and Efficient Structured Generation Engine for LLMs.* MLSys 2025. arXiv:2411.15100 — Up to 100× over prior grammar-constrained decoding.
- **Min, Y. & Azizan, N.** (2024/25). *HardNet: Hard-Constrained Neural Networks with Universal Approximation Guarantees.* arXiv:2410.10807 ⚠ arXiv only as of v4.

- **Banerjee, D., Suresh, T., Ugare, S., Misailovic, S. & Singh, G.** (2025). *CRANE: Reasoning with constrained LLM generation.* ICML 2025. arXiv:2502.09061 — Up to 10 points on GSM-Symbolic and FOLIO by augmenting the grammar with a reasoning region.
- **Kurscheidt, et al.** (2025). *PAL: A Probabilistic Neuro-symbolic Layer for Algebraic Constraint Satisfaction.* UAI 2025. arXiv:2503.19466
- **Kaulen, K., Ladner, T., Bak, S., Brix, C., et al.** (2025). *The 6th International Verification of Neural Networks Competition (VNN-COMP 2025).* arXiv:2512.19007 — α,β-CROWN placed first. ⚠ Detailed scores came from a third-party summary page; team/benchmark counts were corroborated on the arXiv abstract.
- **Pal, S. & Li, C.** (2026). *DisjunctiveNet: Neural Symbolic Learning via Differentiable Convexified Optimization Layers.* ICML 2026 ⚠ (venue from the author-supplied arXiv comment field). arXiv:2605.30456
- ⚠ **Ramirez, Hashemizadeh & Lacoste-Julien.** *Position: Adopt Constraints Over Fixed Penalties in Deep Learning.* arXiv:2505.20628 — **Could not be confirmed to exist during the audit.** Verify before citing.

---

7. Perception, concepts, program induction, abstract reasoning

- **Andreas, J., Rohrbach, M., Darrell, T. & Klein, D.** (2015). *Neural Module Networks.* arXiv:1511.02799 — Direct ancestor of NS-VQA/NS-CL. ⚠ CVPR 2016 venue not verified.
- **Johnson, J., Hariharan, B., van der Maaten, L., Fei-Fei, L., Zitnick, C.L. & Girshick, R.** (2017). *CLEVR.* CVPR 2017. link
- **Yi, K., Wu, J., Gan, C., Torralba, A., Kohli, P. & Tenenbaum, J.B.** (2018). *Neural-Symbolic VQA.* NeurIPS 2018 (spotlight). arXiv:1810.02338 — **99.8% on CLEVR** (verified).
- **Mao, J., Gan, C., Kohli, P., Tenenbaum, J.B. & Wu, J.** (2019). *The Neuro-Symbolic Concept Learner.* ICLR 2019 (Oral). arXiv:1904.12584 ⚠ The widely-quoted 98.9% CLEVR figure does not appear in the abstract; it is unverified.
- **Yi, K., Gan, C., Li, Y., Kohli, P., Wu, J., Torralba, A. & Tenenbaum, J.B.** (2020). *CLEVRER.* ICLR 2020. arXiv:1910.01442 — NS-DR: 88.1% descriptive, 79.6% explanatory, 68.7% predictive, **42.2% counterfactual**.
- **Koh, P.W., Nguyen, T., Tang, Y.S., Mussmann, S., Pierson, E., Kim, B. & Liang, P.** (2020). *Concept Bottleneck Models.* ICML 2020.
- **Locatello, F., Weissenborn, D., Unterthiner, T., Mahendran, A., Heigold, G., Uszkoreit, J., Dosovitskiy, A. & Kipf, T.** (2020). *Object-Centric Learning with Slot Attention.* NeurIPS 2020. proceedings
- **Mahinpei, A., Clark, J., Lage, I., Doshi-Velez, F. & Pan, W.** (2021). *Promises and Pitfalls of Black-Box Concept Learning Models.* arXiv:2106.13314 — Concept leakage. ⚠ Workshop venue unconfirmed.
- **Stammer, W., Schramowski, P. & Kersting, K.** (2021). *Right for the Right Concept.* CVPR 2021. arXiv:2011.12854 — Introduces CLEVR-Hans3/7.
- **Ellis, K., Wong, C., Nye, M., Sablé-Meyer, M., Cary, L., Morales, L., Hewitt, L., Solar-Lezama, A. & Tenenbaum, J.B.** (2021). *DreamCoder.* PLDI 2021. DOI
- **Kleyko, D., Rachkovskij, D.A., Osipov, E. & Rahimi, A.** (2022/23). *A Survey on Hyperdimensional Computing aka Vector Symbolic Architectures,* Parts I & II. ACM Computing Surveys 55(6) art. 130 and 55(9) art. 175. Part I · Part II
- **Hersche, M., Zeqiri, M., Benini, L., Sebastian, A. & Rahimi, A.** (2023). *A neuro-vector-symbolic architecture for solving Raven's progressive matrices.* Nature Machine Intelligence. link — 87.7% RAVEN, 88.1% I-RAVEN.

- **Gupta, T. & Kembhavi, A.** (2023). *Visual Programming.* CVPR 2023 (Best Paper), pp. 14953–14962. Concurrent: ViperGPT (Surís, Menon & Vondrick, ICCV 2023, arXiv:2303.08128) ⚠ *its 48.1% GQA figure was seen only in a third-party paper's description and is unverified.*
- **Grand, G., Wong, L., Bowers, M., Olausson, T.X., Liu, M., Tenenbaum, J.B. & Andreas, J.** (2024). *LILO.* ICLR 2024. arXiv:2310.19791
- **Chollet, F., Knoop, M., Kamradt, G., Landers, B. & Pinkard, H.** (2025). *ARC-AGI-2.* arXiv:2505.11831 ⚠ **No human-baseline percentage appears in this paper or the 2025 technical report — do not quote one.**
- **Jolicoeur-Martineau, A.** (2025). *Less is More: Recursive Reasoning with Tiny Networks* (TRM). arXiv:2510.04871 — 7M parameters, 45% ARC-AGI-1, 8% ARC-AGI-2. ARC Prize 2025 Paper Award, 1st place.
- **Pourcel, Colas & Oudeyer** (2025). *SOAR.* ICML 2025. arXiv:2507.14172 — Evolutionary LLM program synthesis, 52% ARC-AGI-1 public test, no DSL.
- **Chollet, F., Knoop, M., Kamradt, G. & Landers, B.** (2026). *ARC Prize 2025: Technical Report.* arXiv:2601.10904 — 1,455 teams, 15,154 entries, top private-eval **24%**, 90 paper submissions. ⚠ Per-team names and $/task figures are single-source (arcprize.org blog) and the two sources disagree on cost.
- **Yang, Y., Campbell, D., Huang, K., Wang, M., Cohen, J. & Webb, T.** (2025). *Emergent Symbolic Mechanisms Support Abstract Reasoning in Large Language Models.* ICML 2025. arXiv:2502.20332 — Symbol-abstraction heads → symbolic-induction heads → retrieval heads. **The strongest positive evidence that transformers implement symbol-like variable binding internally** — the empirical continuation of Smolensky's programme.

---

8. Knowledge graphs, rules, GNN expressiveness

- **Bordes, A., Usunier, N., García-Durán, A., Weston, J. & Yakhnenko, O.** (2013). *TransE.* NIPS 2013.
- **Yang, B., Yih, W., He, X., Gao, J. & Deng, L.** (2015). *DistMult.* ICLR 2015. arXiv:1412.6575
- **Trouillon, T., Welbl, J., Riedel, S., Gaussier, É. & Bouchard, G.** (2016). *Complex Embeddings for Simple Link Prediction.* ICML 2016. arXiv:1606.06357
- **Sun, Z., Deng, Z.-H., Nie, J.-Y. & Tang, J.** (2019). *RotatE.* ICLR 2019. arXiv:1902.10197
- **Galárraga, L., Teflioudi, C., Hose, K. & Suchanek, F.M.** (2015). *Fast rule mining in ontological knowledge bases with AMIE+.* VLDB Journal. DOI
- **Gutiérrez-Basulto, V. & Schockaert, S.** (2018). *From Knowledge Graph Embedding to Ontology Embedding?* KR 2018. ⚠ Description ("point vs region semantics") from background, not a fetched abstract.
- **Meilicke, C., Chekol, M.W., Ruffinelli, D. & Stuckenschmidt, H.** (2019). *AnyBURL.* IJCAI 2019. DOI; VLDB Journal 33(1):131–161, 2024. Companion: Meilicke et al., *Fine-Grained Evaluation of Rule- and Embedding-Based Systems...* ISWC 2018. ⚠ No specific MRR/Hits@10 numbers verified; the qualitative competitiveness claim is well established.
- **Kulmanov, M., Liu-Wei, W., Yan, Y. & Hoehndorf, R.** (2019). *EL Embeddings: Geometric construction of models for the Description Logic EL++.* arXiv:1902.10499 ⚠ IJCAI-2019 venue unverified. Successors: Box^2EL (arXiv:2301.11118), TransBox, ALC saturation embeddings.
- **Barceló, P., Kostylev, E.V., Monet, M., Pérez, J., Reutter, J.L. & Silva, J.P.** (2020). *The Logical Expressiveness of Graph Neural Networks.* ICLR 2020. OpenReview ⚠ The graded-modal-logic / FOC2 characterisation is from background; OpenReview served a bot-check during verification.

- **Ren, H., Hu, W. & Leskovec, J.** (2020). *Query2box.* ICLR 2020. arXiv:2002.05969; BetaE, NeurIPS 2020, arXiv:2010.11465
- **Zhu, Z., Zhang, Z., Xhonneux, L.-P. & Tang, J.** (2021). *Neural Bellman-Ford Networks.* NeurIPS 2021. arXiv:2106.06935 — **The conceptual pivot of the sub-area.**
- **Veličković, P. & Blundell, C.** (2021). *Neural Algorithmic Reasoning.* Patterns. arXiv:2105.02761; CLRS benchmark, ICML 2022, arXiv:2205.15659
- **Grohe, M.** (2023/24). *The Descriptive Complexity of Graph Neural Networks.* TheoretiCS vol. 3. arXiv:2303.04613; readable on-ramp: *The Logic of Graph Neural Networks,* arXiv:2104.14624
- **Zhu, Z., Yuan, X., Galkin, M., Xhonneux, S., Zhang, M., Gazeau, M. & Tang, J.** (2023). *A*Net.* NeurIPS 2023. arXiv:2206.04798
- **Galkin, M., Yuan, X., Mostafa, H., Tang, J. & Zhu, Z.** (2024). *Towards Foundation Models for Knowledge Graph Reasoning* (ULTRA). ICLR 2024. arXiv:2310.04562 — 57 KGs zero-shot. Extension: UltraQuery, NeurIPS 2024, arXiv:2404.07198
- **Pan, S., Luo, L., Wang, Y., Chen, C., Wang, J. & Wu, X.** (2024). *Unifying Large Language Models and Knowledge Graphs: A Roadmap.* IEEE TKDE. arXiv:2306.08302
- **Edge, D., Trinh, H., Cheng, N., Bradley, J., Chao, A., Mody, A., Truitt, S., Metropolitansky, D., Ness, R.O. & Larson, J.** (2024). *From Local to Global: A Graph RAG Approach…* arXiv:2404.16130 — Read precisely to see what it is **not**: the graph is an index; no inference, no entailment, no consistency check.
- **Huang, X., Barceló, P., Bronstein, M.M., Ceylan, İ.İ., Galkin, M., Reutter, J.L. & Romero Orth, M.** (2025). *How Expressive are Knowledge Graph Foundation Models?* ICML 2025. arXiv:2502.13339
- **Fan, D., Xue, Z., Liu, S. & Tan, Q.** (2026). *Do We Still Need GraphRAG? Benchmarking RAG and GraphRAG for Agentic Search Systems.* arXiv:2604.09666

---

9. LLM + solver / prover

- **Gao, L., Madaan, A., Zhou, S., et al.** (2022). *PAL: Program-aided Language Models.* arXiv:2211.10435 ⚠ ICML 2023 venue believed but unconfirmed. Concurrent twin: Program-of-Thoughts (Chen et al., TMLR 2023, arXiv:2211.12588).
- **Lyu, Q., Havaldar, S., Stein, A., et al.** (2023). *Faithful Chain-of-Thought Reasoning.* IJCNLP-AACL 2023. arXiv:2301.13379
- **Ye, X., Chen, Q., Dillig, I. & Durrett, G.** (2023). *SatLM.* NeurIPS 2023. arXiv:2305.09656
- **Pan, L., Albalak, A., Wang, X. & Wang, W.Y.** (2023). *Logic-LM.* Findings of EMNLP 2023. arXiv:2305.12295 — +39.2% over standard prompting, +18.4% over CoT.
- **Olausson, T.X., Gu, A., Lipkin, B., et al.** (2023). *LINC.* EMNLP 2023 (Outstanding Paper). arXiv:2310.15164
- **Yang, Z., Ishay, A. & Lee, J.** (2023). *Coupling Large Language Models with Logic Programming…* Findings of ACL 2023, pp. 5186–5219. arXiv:2307.07696; companion KR 2023, arXiv:2307.07699
- **Yang, K., Swope, A.M., Gu, A., Chalamala, R., Song, P., Yu, S., Godil, S., Prenger, R. & Anandkumar, A.** (2023). *LeanDojo.* NeurIPS 2023 D&B (oral). arXiv:2306.15626 — 98,734 theorems from mathlib; turns Lean into a Python-callable gym.
- **Trinh, T.H., Wu, Y., Le, Q.V., He, H. & Luong, T.** (2024). *Solving olympiad geometry without human demonstrations* (AlphaGeometry). Nature 625(7995):476–482. DOI — **25/30 vs prior best 10.** Fully confirmed against the PMC full text. AlphaGeometry 2: arXiv:2502.03544.

- **Kambhampati, S., Valmeekam, K., Guan, L., et al.** (2024). *LLMs Can't Plan, But Can Help Planning in LLM-Modulo Frameworks.* ICML 2024, PMLR 235. arXiv:2402.01817
- **Mirzadeh, I., Alizadeh, K., Shahrokhi, H., Tuzel, O., Bengio, S. & Farajtabar, M.** (2024/25). *GSM-Symbolic.* ICLR 2025. arXiv:2410.05229 — Up to 65% degradation from one logically inert clause. Full six-author list confirmed.
- **Lam, L.H.M., Thatikonda, R.K. & Shareghi, E.** (2024). *A Closer Look at Logical Reasoning with LLMs: The Choice of Tool Matters.* arXiv:2406.00284 — **~50% of reported variation traces to solver choice.**
- **DeepSeek-AI (Guo, D., Yang, D., Zhang, H., Song, J., Wang, P., et al.)** (2025). *DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning.* Nature 645(8081):633–638. DOI *[corrected]* This is the **Nature** title; "DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning" is the arXiv:2501.12948 title. ⚠ The widely quoted 79.8% AIME 2024 pass@1 was not verified.
- **Shojaee, P., Mirzadeh, I., Alizadeh, K., Horton, M., Bengio, S. & Farajtabar, M.** (2025). *The Illusion of Thinking.* NeurIPS 2025. arXiv:2506.06941
- **Lawsen, A.** (2025). *Comment on The Illusion of Thinking.* arXiv:2506.09250 — The methodological counterweight. Two further rebuttals followed within a month (2506.18957, 2507.01231).
- **Ren, Z.Z., Shao, Z., Song, J., et al. (DeepSeek-AI)** (2025). *DeepSeek-Prover-V2.* arXiv:2504.21801 — 88.9% miniF2F-test, 49/658 PutnamBench. Lineage: V1.5 (2408.08152), Goedel-Prover (2502.07640), Kimina-Prover (2504.11354).
- **Chen, L., Gu, J., Huang, L., et al. (ByteDance Seed)** (2025). *Seed-Prover.* arXiv:2507.23726 — **5/6 IMO 2025 problems formally proved in Lean.**
- **Hubert, T., Mehta, R., Sartran, L., … Hassabis, D., Kohli, P. & Silver, D.** (2025). *Olympiad-level formal mathematical reasoning with reinforcement learning* (AlphaProof). Nature **651(8106):607–613**, epub 12 Nov 2025. DOI — Fully confirmed via PubMed 41225005. 3/5 non-geometry IMO 2024 problems; 28 points with AlphaGeometry 2 = silver. *[corrected]* Cite as **2025** (epub); the print issue is March 2026 — this is why the year appears inconsistently across sources.
- **Chen, J., Chen, W., Du, J., et al. (ByteDance Seed)** (2025). *Seed-Prover 1.5.* arXiv:2512.17260 *[corrected]* The abstract states **88%** of PutnamBench — the circulating "87.9%" does not appear there.
- **Chung, J.-H., Cai, Z., Li, Z., … Wang, M., Chen, D., Jin, C., Fowl, L.H. & Arora, S.** (2026). *Goedel-Architect.* arXiv:2606.06468 — 99.2% pass@1 miniF2F-test, 75.6% pass@1 PutnamBench; 100% / 88.8% (597/672) with NL seeding; 4/6 IMO 2025, 11/12 Putnam 2025, 3/6 USAMO 2026, at up to 500× lower cost. **All figures confirmed verbatim in the audit.**
- **Raiyan, S.R., Kabir, M., Mahmud, H., Hasan, M.K. & Ananiadou, S.** (2026). *Artificial Intelligence for Mathematical Reasoning: An Integrated Survey…* arXiv:2606.08728 — 47 pages; unusually honest about benchmark saturation, contamination and pass@k reporting mismatches.
- **Romera-Paredes, B., Barekatain, M., Novikov, A., Balog, M., Kumar, M.P., Dupont, E., Ruiz, F.J.R., Ellenberg, J.S., Wang, P., Fawzi, O., Kohli, P. & Fawzi, A.** (2024). *Mathematical discoveries from program search with large language models* (FunSearch). Nature 625(7995):468–475. link — The canonical LLM-plus-verifier discovery result and the direct predecessor of AlphaEvolve.
- **Novikov, A., Vũ, N., Eisenberger, M., Dupont, E., Huang, P.-S., Wagner, A.Z., et al.** (2025). *AlphaEvolve.* arXiv:2506.13131 — 48-multiplication 4×4 complex matrix multiplication, first improvement over Strassen in 56 years.
- **Georgiev, B., Gómez-Serrano, J., Tao, T. & Wagner, A.Z.** (2025). *Mathematical exploration and discovery at scale.* arXiv:2511.02864 — 67 open/semi-open problems.

- **Kordjamshidi, P., Aslan, S., Seshadri, M., Barrett, L. & Santus, E.** (2026). *Reasoners or Translators? Contamination-aware Evaluation and Neuro-Symbolic Robustness in Tax Law.* arXiv:2605.16052

⚠ **Unverified during the audit and excluded from the review's claims:** Harmonic AI's "Aristotle" IMO 2025 gold result (no matching paper found); the OpenAI July 2025 IMO 35/42 claim (no fetchable primary source); a cluster of 2026 arXiv IDs surfaced only in search listings (2606.16541, 2607.19407, 2511.03108, 2601.14456, 2606.12594, 2605.27014, 2606.16603, 2607.23386). Re-check any of these before citing.

---

10. Reasoning shortcuts and evaluation critique

- **Marconato, E., Teso, S., Vergari, A. & Passerini, A.** (2023). *Not All Neuro-Symbolic Concepts Are Created Equal.* NeurIPS 2023. arXiv:2305.19951
- **Bortolotti, S., Marconato, E., Carraro, T., Morettin, P., van Krieken, E., Vergari, A., Teso, S. & Passerini, A.** (2024). *A Neuro-Symbolic Benchmark Suite for Concept Quality and Reasoning Shortcuts* (rsbench). arXiv:2406.10368 · site — miniBOIA: DeepProbLog 0.87 label / **0.28 concept**; LTN 0.78/**0.35**; CLIP 0.99/**0.34**. ⚠ NeurIPS 2024 D&B venue could not be confirmed from arXiv metadata; the eight-author list is confirmed verbatim.
- **van Krieken, E., Minervini, P., Ponti, E.M. & Vergari, A.** (2025). *Neurosymbolic Reasoning Shortcuts under the Independence Assumption.* NeSy 2025. arXiv:2507.11357
- **Marconato, E., Bortolotti, S., van Krieken, E., Morettin, P., Umili, E., Vergari, A., Tsamoura, E., Passerini, A. & Teso, S.** (2025/26). *Symbol Grounding in Neuro-Symbolic AI: A Gentle Introduction to Reasoning Shortcuts.* JAIR (special track). arXiv:2510.14538 (v3, Aug 2026) — Nine-author list and JAIR status confirmed. ⚠ No volume/pages assigned.
- **Takemura, A., Inoue, K. & Nishino, M.** (2026). *Constraint-Based Analysis of Reasoning Shortcuts in Neurosymbolic Learning.* KR 2026. arXiv:2604.23377 — Shortcut detection coNP-complete, counting #P-complete. ⚠ Verified via abstract page only; the arXiv ID should be re-checked before citation.
- **Lake, B.M. & Baroni, M.** (2018). *Generalization without systematicity* (SCAN). ICML 2018. arXiv:1711.00350
- **Zhou, D., Schärli, N., Hou, L., Wei, J., Scales, N., Wang, X., Schuurmans, D., Cui, C., Bousquet, O., Le, Q. & Chi, E.** (2022/23). *Least-to-Most Prompting.* ICLR 2023. arXiv:2205.10625 **— ≥99% on all SCAN splits with 14 exemplars vs 16% CoT.**
- **Drozdov, A., Schärli, N., Akyürek, E., Scales, N., Song, X., Chen, X., Bousquet, O. & Zhou, D.** (2022). *Compositional Semantic Parsing with Large Language Models.* arXiv:2209.15003 — CFQ 95.0 mean with **1%** of the training data. ⚠ ICLR 2023 venue unconfirmed; author list confirmed verbatim.
- **Lake, B.M. & Baroni, M.** (2023). *Human-like systematic generalization through a meta-learning neural network.* Nature 623(7985):115–121. DOI — **The most direct empirical rebuttal of Fodor & Pylyshyn available.**
- **Turpin, M., Michael, J., Perez, E. & Bowman, S.R.** (2023). *Language Models Don't Always Say What They Think.* NeurIPS 2023. Companion: Lanham et al., arXiv:2307.13702.
- **Ott, Ledaguenel, Hudelot & Hartwig** (2023). *How to Think About Benchmarking Neurosymbolic AI?* NeSy 2023, CEUR-WS Vol-3432, pp. 248–254. ⚠ PDF text not extractable; arguments not characterised here.
- **Chan, Gaizauskas & Zhao** (2026). *Position: Logical Soundness is not a Reliable Criterion for Neurosymbolic Fact-Checking with LLMs.* ICLR 2026 workshop. arXiv:2604.04177

---

11. RL, planning, world models

- **Toro Icarte, R., Klassen, T.Q., Valenzano, R. & McIlraith, S.A.** (2018). *Using Reward Machines for High-Level Task Specification and Decomposition in RL.* ICML 2018, PMLR 80:2107–2116. link
- **Toro Icarte, R., Klassen, T.Q., Valenzano, R. & McIlraith, S.A.** (2022). *Reward Machines: Exploiting Reward Function Structure in RL.* JAIR 73:173–208. arXiv:2010.03950 — **The best self-contained entry point to neurosymbolic RL.**
- **Verma, A., Murali, V., Singh, R., Kohli, P. & Chaudhuri, S.** (2018). *Programmatically Interpretable Reinforcement Learning.* ICML 2018, PMLR 80:5045–5054. arXiv:1804.02477
- **Bastani, O., Pu, Y. & Solar-Lezama, A.** (2018). *Verifiable RL via Policy Extraction* (VIPER). arXiv:1805.08328 ⚠ NeurIPS 2018 venue not confirmed from the arXiv page.
- **De Giacomo, G., Iocchi, L., Favorito, M. & Patrizi, F.** (2019). *Foundations for Restraining Bolts.* ICAPS 2019, pp. 128–136. arXiv:1807.06333
- **Dong, H., Mao, J., Lin, T., Wang, C., Li, L. & Zhou, D.** (2019). *Neural Logic Machines.* ICLR 2019 ⚠ (venue from arXiv comments).
- **Jiang, Z. & Luo, S.** (2019). *Neural Logic Reinforcement Learning.* ICML 2019 ⚠ (venue from arXiv comments). arXiv:1904.10729
- **Jansen, N., Könighofer, B., Junges, S., Serban, A. & Bloem, R.** (2020). *Safe Reinforcement Learning Using Probabilistic Shields (Invited Paper).* CONCUR 2020, pp. 3:1–3:16, DOI 10.4230/LIPIcs.CONCUR.2020.3. *[corrected]* The arXiv version (1807.06096) is titled "...via Probabilistic Shields"; cite the CONCUR record.
- **Vaezipoor, P., Li, A., Toro Icarte, R. & McIlraith, S.** (2021). *LTL2Action.* ICML 2021.
- **Ahn, M., Brohan, A., Brown, N., et al.** (2022). *Do As I Can, Not As I Say* (SayCan). CoRL 2022, pp. 287–318. arXiv:2204.01691 — 101 instructions, 84% plan / 74% execution on a real robot. ⚠ Figures from the project page, not the proceedings PDF.
- **Liang, J., Huang, W., Xia, F., Xu, P., Hausman, K., Ichter, B., Florence, P. & Zeng, A.** (2023). *Code as Policies.* ICRA 2023, pp. 9493–9500. arXiv:2209.07753
- **Valmeekam, K., Marquez, M., Olmo, A., Sreedharan, S. & Kambhampati, S.** (2023). *PlanBench.* NeurIPS 2023 D&B. arXiv:2206.10498 — The obfuscated "Mystery" variants are the key device.
- **Guan, L., Valmeekam, K., Sreedharan, S. & Kambhampati, S.** (2023). *Leveraging Pre-trained LLMs to Construct and Utilize World Models for Model-based Task Planning.* NeurIPS 2023. arXiv:2305.14909
- **Acharya, K., Raza, W., Dourado Jr., C.M.J.M., Velasquez, A. & Song, H.H.** (2023/24). *Neurosymbolic Reinforcement Learning and Planning: A Survey.* IEEE TAI 5(5):1939–1953. arXiv:2309.01038 — Use as a map, not an authority on results.
- **Valmeekam, K., Stechly, K. & Kambhampati, S.** (2024). *LLMs Still Can't Plan; Can LRMs?* arXiv:2409.13373 — o1-preview 97.8% Blocksworld / 37.3% randomised Mystery; Fast Downward 100% at 0.265 s. ⚠ Numbers from arXiv HTML v1, not cross-checked against a second source.
- **Liang, Y., Kumar, N., Tang, H., Weller, A., Tenenbaum, J.B., Silver, T., Henriques, J.F. & Ellis, K.** (2025). *VisualPredicator.* ICLR 2025 (Spotlight). arXiv:2410.23156
- **Khan, Z., Prasad, A., Stengel-Eskin, E., Cho, J. & Bansal, M.** (2026). *One Life to Learn* (OneLife). ICLR 2026. arXiv:2510.12088
- **Lorang, P., Huemer, J., Duggan, T., Goebel, K., Zips, P. & Scheutz, M.** (2026). *Build on Priors.* arXiv:2604.03759 — Real industrial forklift and Kinova arm; PDDL domain synthesised via ASP from 1–30 unannotated demonstrations. ⚠ arXiv preprint, no venue.

12. Applications (domains the core NeSy literature under-covers)

⚠ *These were surfaced by a completeness sweep and verified as existing works, but their results were not independently audited. Treat as entry points, not as endorsements.*

- **Cybersecurity** — Hakim, S.B., Adil, M., Velasquez, A., Xu, S. & Song, H.H. (2025). *Neuro-Symbolic AI for Cybersecurity: State of the Art, Challenges, and Opportunities.* arXiv:2509.06921. Companions: Tran et al. (2506.04454), Samaddar et al. on OOD detection for neurosymbolic cyber agents (2412.02875). **This is where DARPA/ARL funding actually lands.**
- **Drug discovery** — MARS, arXiv:2410.05289
- **Bioinformatics / ontologies** — the EL Embeddings line (§8), Hoehndorf's KAUST group
- **Scientific discovery** — AI Feynman (Science Advances 6:eaay2631); FunSearch (Nature 625:468–475); AlphaEvolve (§9)
- **Manufacturing** — CausalTrace, AAAI-26 IAAI, arXiv:2510.12033; CausalPulse ⚠ (Bosch-plant deployment claim from a search summary)
- **Clinical, legal, finance, EDA** — several 2026 arXiv entries were surfaced but could not be individually verified; they are deliberately not listed with IDs here.

---

13. Venues, institutions, community

- **NeSy** — International Conference on Neurosymbolic Learning and Reasoning. 19th edition Sep 2025 (UC Santa Cruz, PMLR vol. 284, eds. Gilpin, Giunchiglia, Hitzler & van Krieken; keynotes Van den Broeck, Kipf, McGuinness); 20th edition Lisbon (FCUL), 1–4 Sep 2026. Began 2005 as a workshop series; **not annual** (annual since 2005 would make 2025 the 21st). Run by the Neurosymbolic AI Association (nesy-ai.org), formalised at a 2014 Dagstuhl seminar.
- **IJCLR** — 5th edition Surrey Sep 2025, 6th UPV Valencia 16–18 Sep 2026. Inductive-logic-programming leaning.
- **NeuS** — International Conference on Neuro-symbolic Systems, 2026 edition. Usually missing from venue lists.
- ***Neurosymbolic Artificial Intelligence*** journal — IOS Press / SAGE, ISSN 2949-8732, open access, continuous publication. ⚠ Sources disagree on launch year; the homepage shows articles from 2024–25.
- **Adjacent venues** — KR, ICLP, AAAI, IJCAI, NeurIPS/ICML workshops. Note: **NeurIPS 2025 had no neurosymbolic-branded workshop**; the reasoning workshops are LLM-centric (MATH-AI, Foundations of Reasoning in Language Models, Efficient Reasoning).
- **IJCAI-ECAI 2026** (Bremen, 15–17 Aug) tutorial T21, *Deep Parameterized Logics as A Foundation for Neurosymbolic AI* — De Raedt, Marra, Manhaeve & Derkinderen.
- **Summer schools** — ScaDS.AI School on Neuro+Symbolic AI, Leipzig, 22–26 June 2026 (speakers incl. Belle, Hitzler, Ilievski, Marra) — verified. ⚠ A Centaur AI Institute 2026 edition surfaced in search but its page 404'd.
- **DARPA ANSR** — Assured Neuro Symbolic Learning and Reasoning, PM Susmit Jha, IPTO, BAA HR001122S0039. ⚠ No 2025–26 status update, budget or performer list on the programme page.

Groups to follow — KU Leuven (De Raedt, Manhaeve, Marra: DeepProbLog/DeepLog) · UCLA StarAI (Van den Broeck: circuits, semantic loss) · Edinburgh (Vergari, Minervini, van Krieken) · Trento (Teso, Passerini, Marconato: reasoning shortcuts) · VU Amsterdam (van Harmelen) · UPenn (Naik, Li: Scallop/Dolphin) · City, University of London (d'Avila Garcez) · Kansas State (Hitzler) · Siena (Gori, Diligenti, Melacci) · MIT/Cornell (Tenenbaum, Ellis: program induction) · **Nanjing LAMDA (Zhou, Dai: abductive learning) · NII Tokyo (Inoue, Sato: differentiable ASP)** · Arizona State (Kambhampati: planning critique; Shakarian: PyReason) · IBM Research (LNN) · Google DeepMind (Alpha- *systems). ⚠ Individual affiliations shift; several could not be re-verified during preparation. Check before citing anyone's institution.*